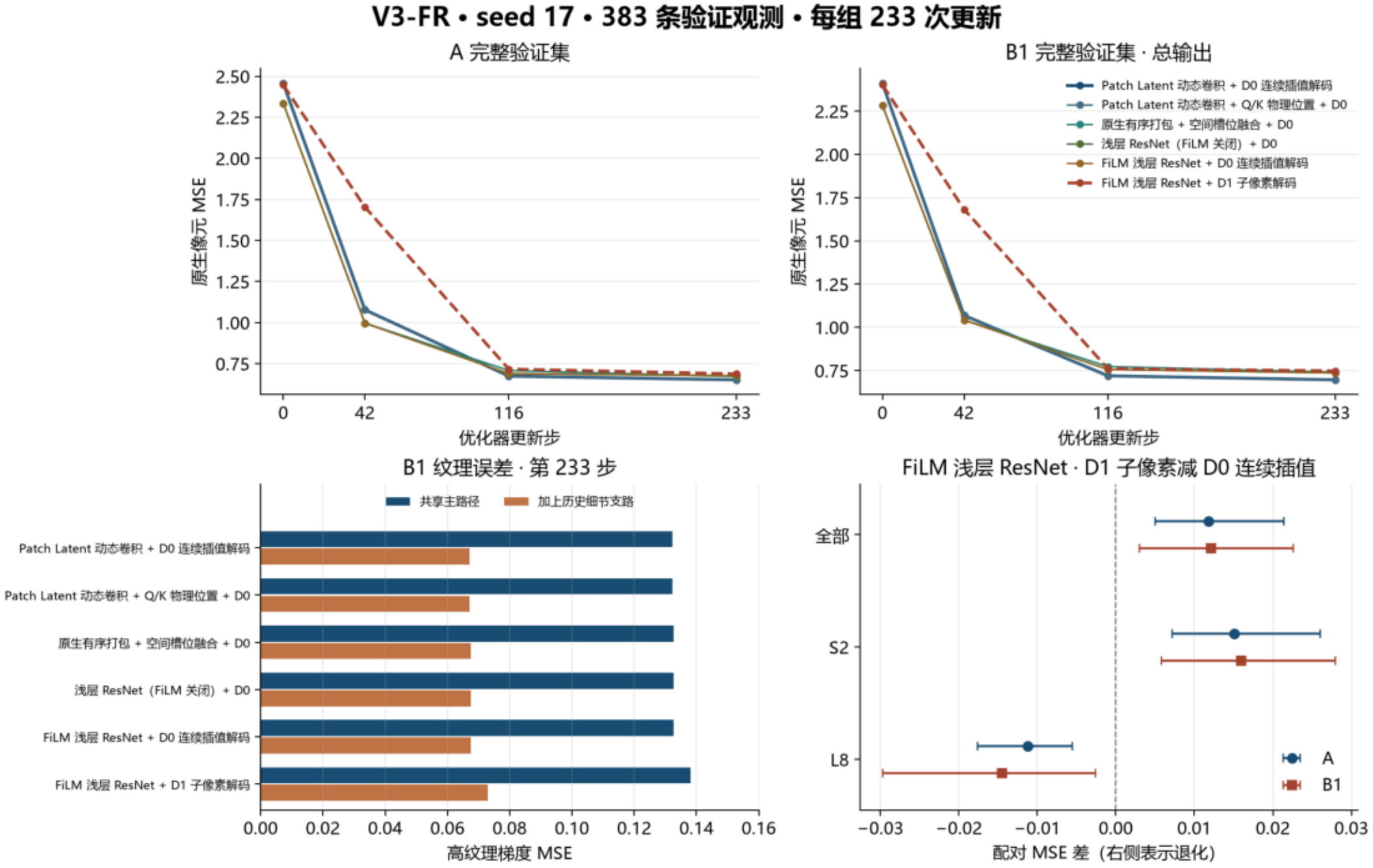
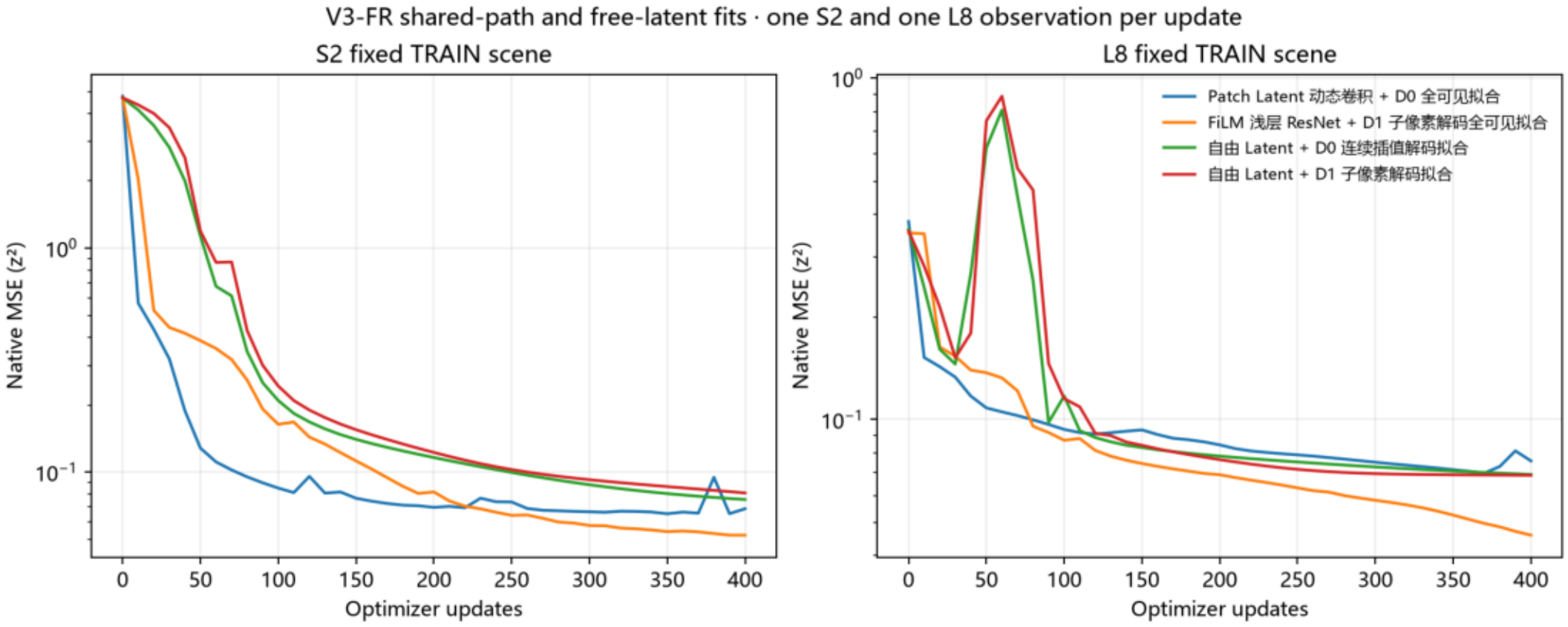
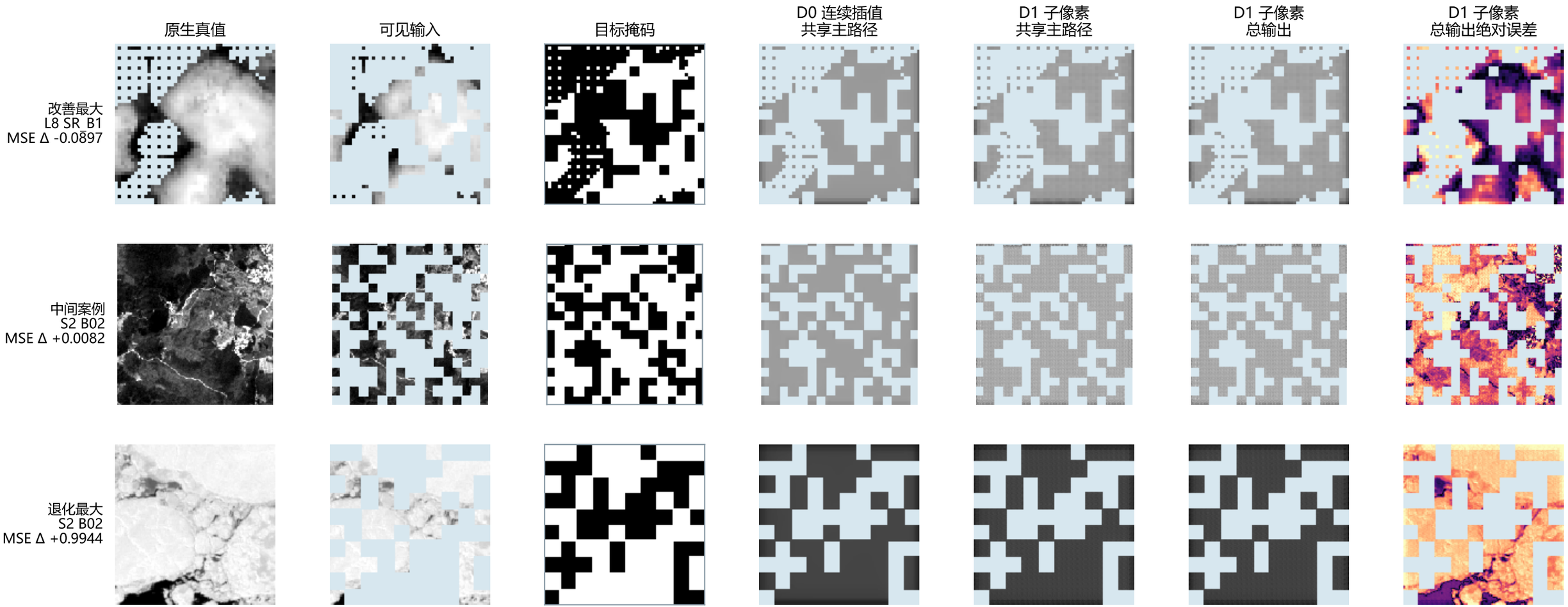
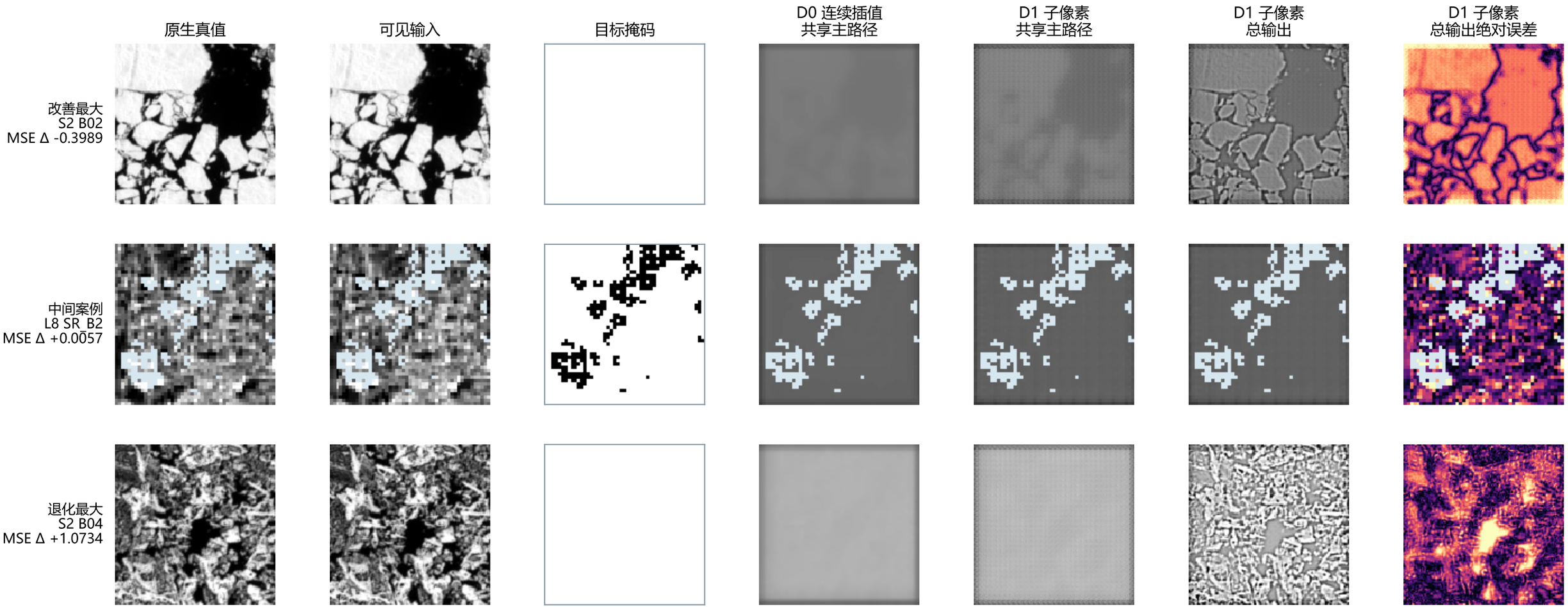

V3-FR • seed 17 • 383 条验证观测 • 每组 233 次更新
A 完整验证集
B1 完整验证集 · 总输出
原生像元 MSE
优化器更新步
Patch Latent 动态卷积 + D0 连续插值解码
Patch Latent 动态卷积 + Q/K 物理位置 + D0
原生有序打包 + 空间槽位融合 + D0
浅层 ResNet（FiLM 关闭） + D0
FiLM 浅层 ResNet + D0 连续插值解码
FiLM 浅层 ResNet + D1 子像素解码
B1 纹理误差 · 第 233 步
共享主路径
加上历史细节支路
高纹理梯度 MSE
FiLM 浅层 ResNet · D1 子像素减 D0 连续插值
全部
S2
L8
A
B1
配对 MSE 差（右侧表示退化）


V3-FR shared-path and free-latent fits · one S2 and one L8 observation per update
S2 fixed TRAIN scene
L8 fixed TRAIN scene
Native MSE (z²)
Optimizer updates
Patch Latent 动态卷积 + D0 全可见拟合
FiLM 浅层 ResNet + D1 子像素解码全可见拟合
自由 Latent + D0 连续插值解码拟合
自由 Latent + D1 子像素解码拟合

FiLM 浅层 ResNet · D0 连续插值与 D1 子像素解码 · A · 第 233 步



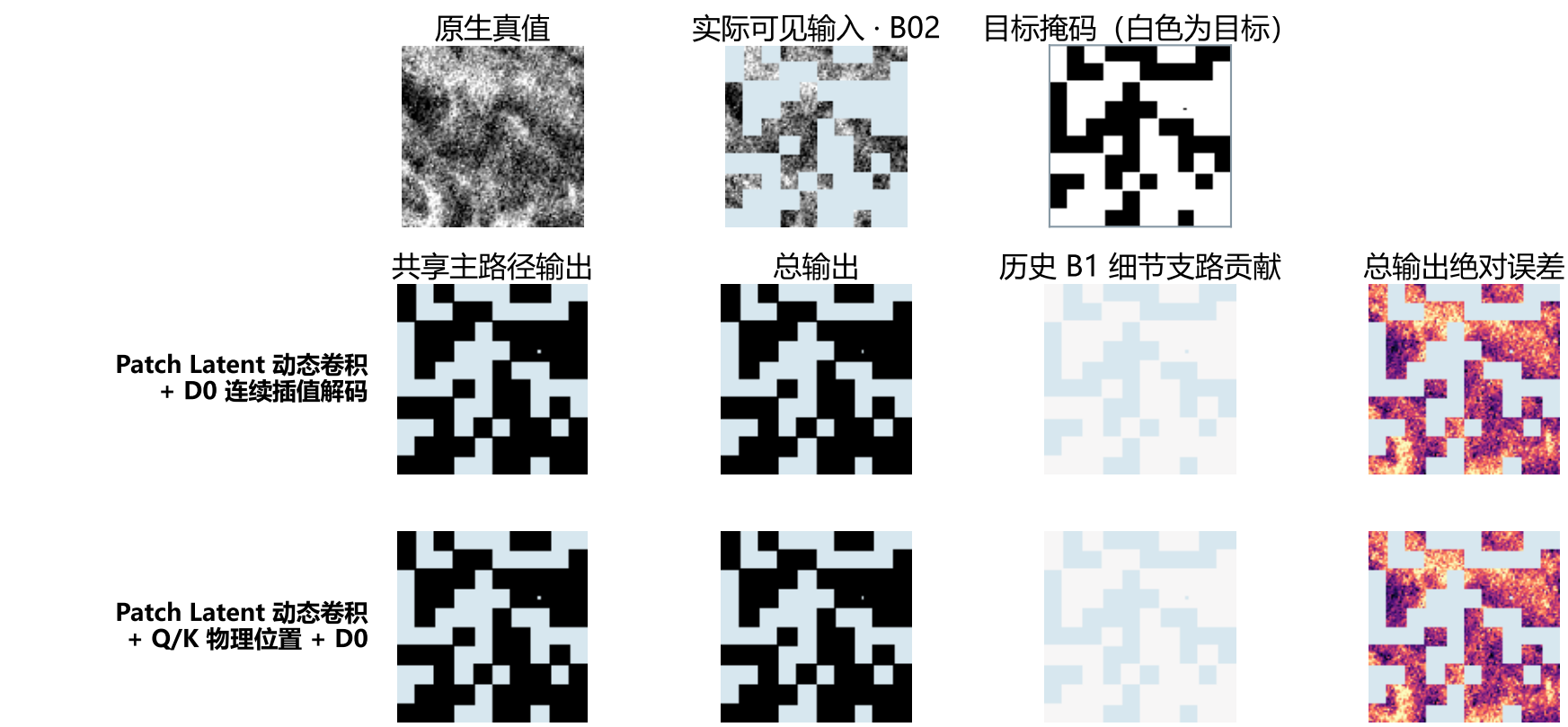
FiLM 浅层 ResNet · D0 连续插值与 D1 子像素解码 · B1 · 第 233 步



S6 • A • S2 • B02 • 第 233 步

obs_5929ceec240fa570c7a1a784

六种模型架构 · 同观测、同目标掩码、同真值色标

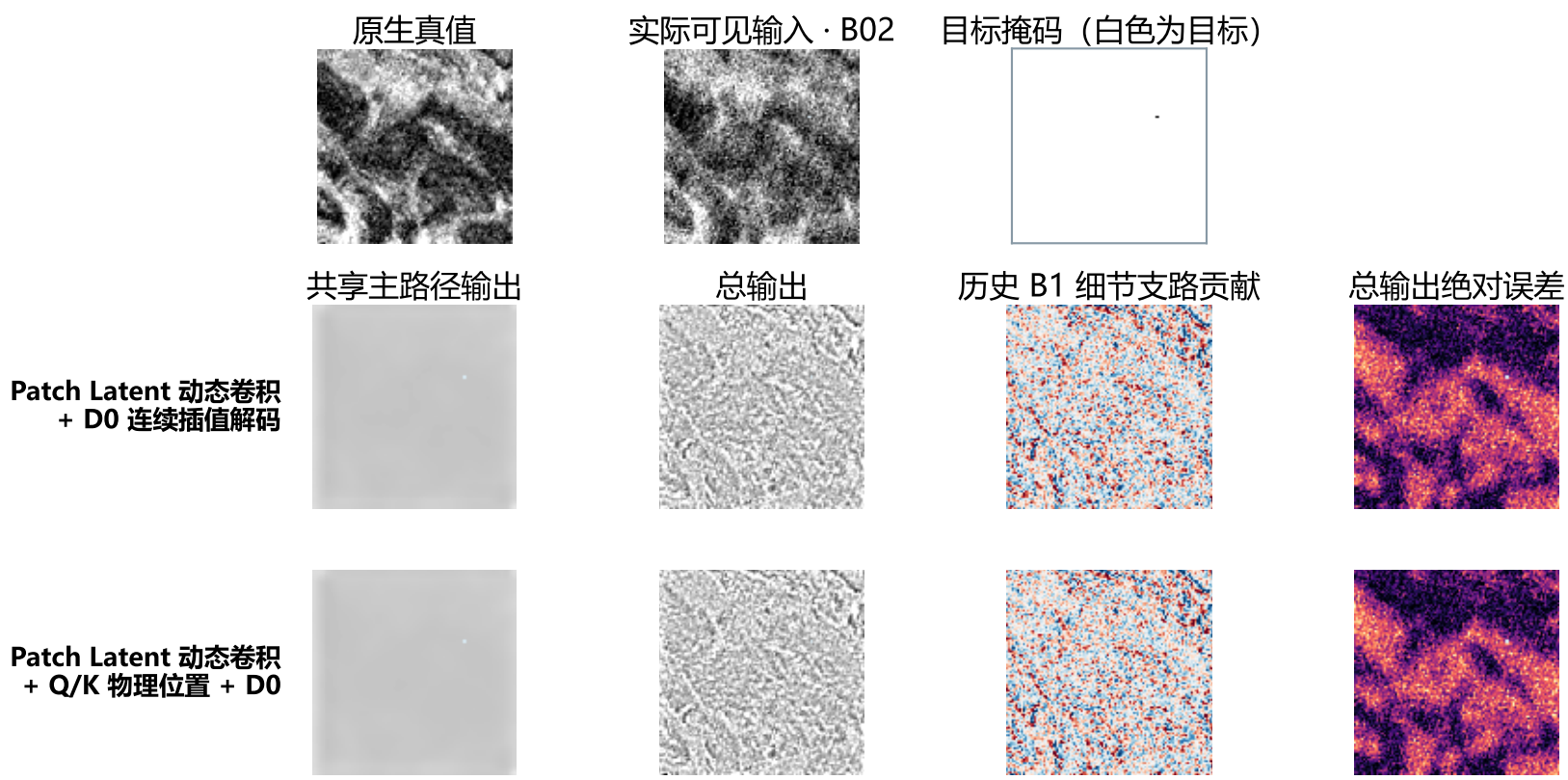


该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S6 • B1 • S2 • B03 • 第 233 步

obs_5929ceec240fa570c7a1a784

六种模型架构 · 同观测、同目标掩码、同真值色标

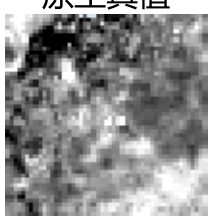
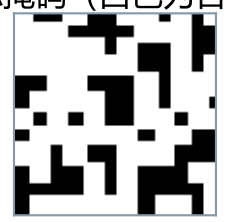
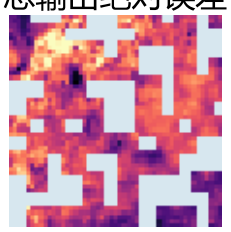
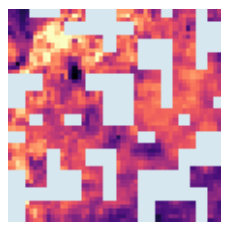
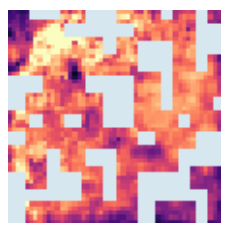
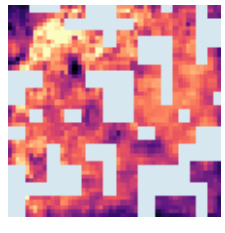
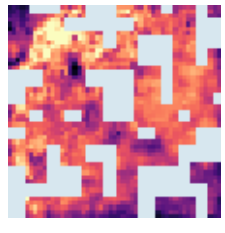
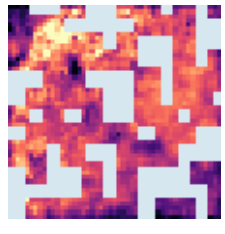


该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S6 • A • L8 • SR_B1 • 第 233 步

obs_64d10dd286169cf55d8668ac

六种模型架构 · 同观测、同目标掩码、同真值色标

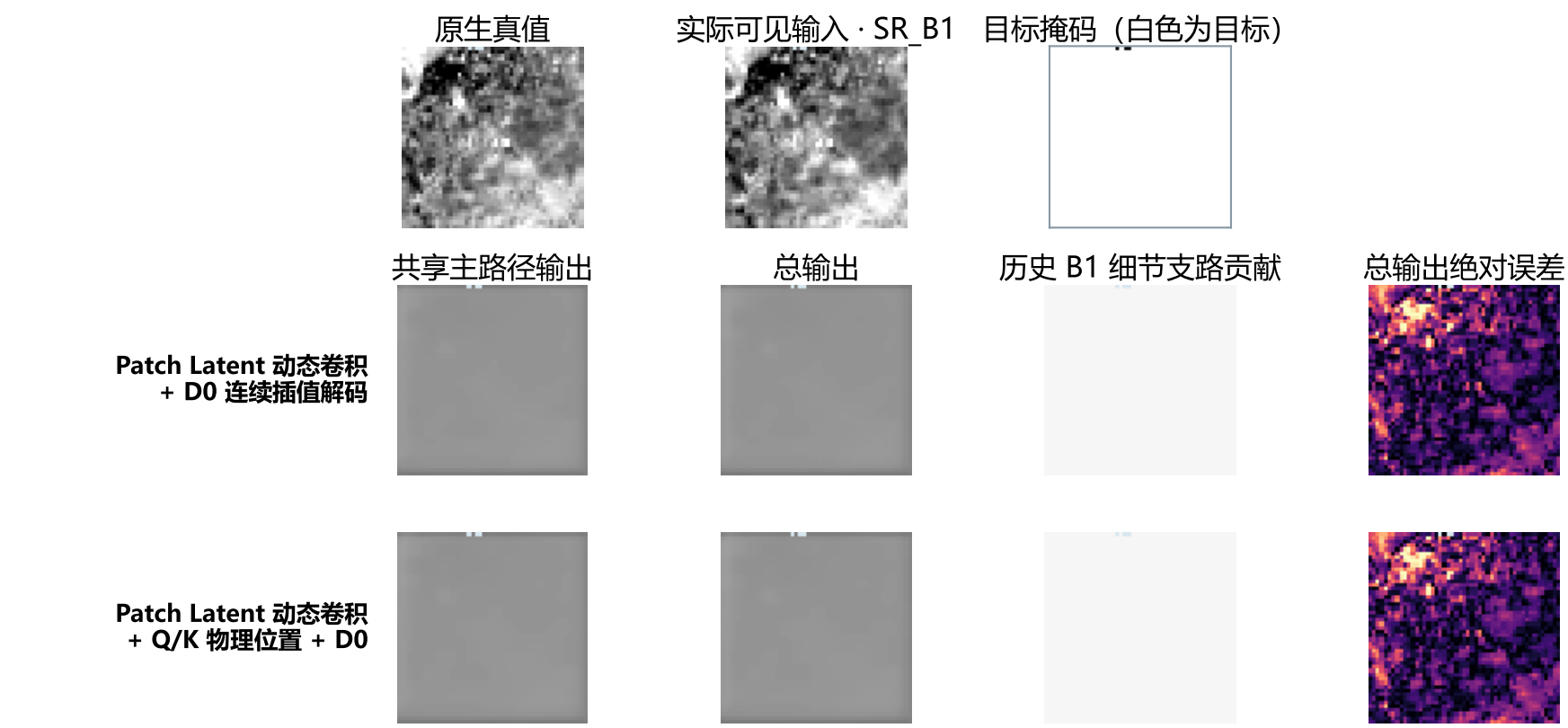
	原生真值	实际可见输入 · SR_B1	目标掩码（白色为目标）		
					
	共享主路径输出	总输出	历史 B1 细节支路贡献	总输出绝对误差	
Patch Latent 动态卷积 + D0 连续插值解码					
Patch Latent 动态卷积 + Q/K 物理位置 + D0					
原生有序打包 + 空间槽位融合 + D0					
浅层 ResNet（FiLM 关闭）+ D0					
FiLM 浅层 ResNet + D0 连续插值解码					
FiLM 浅层 ResNet + D1 子像素解码					

该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S6 • B1 • L8 • SR_B2 • 第 233 步

obs_64d10dd286169cf55d8668ac

六种模型架构 · 同观测、同目标掩码、同真值色标

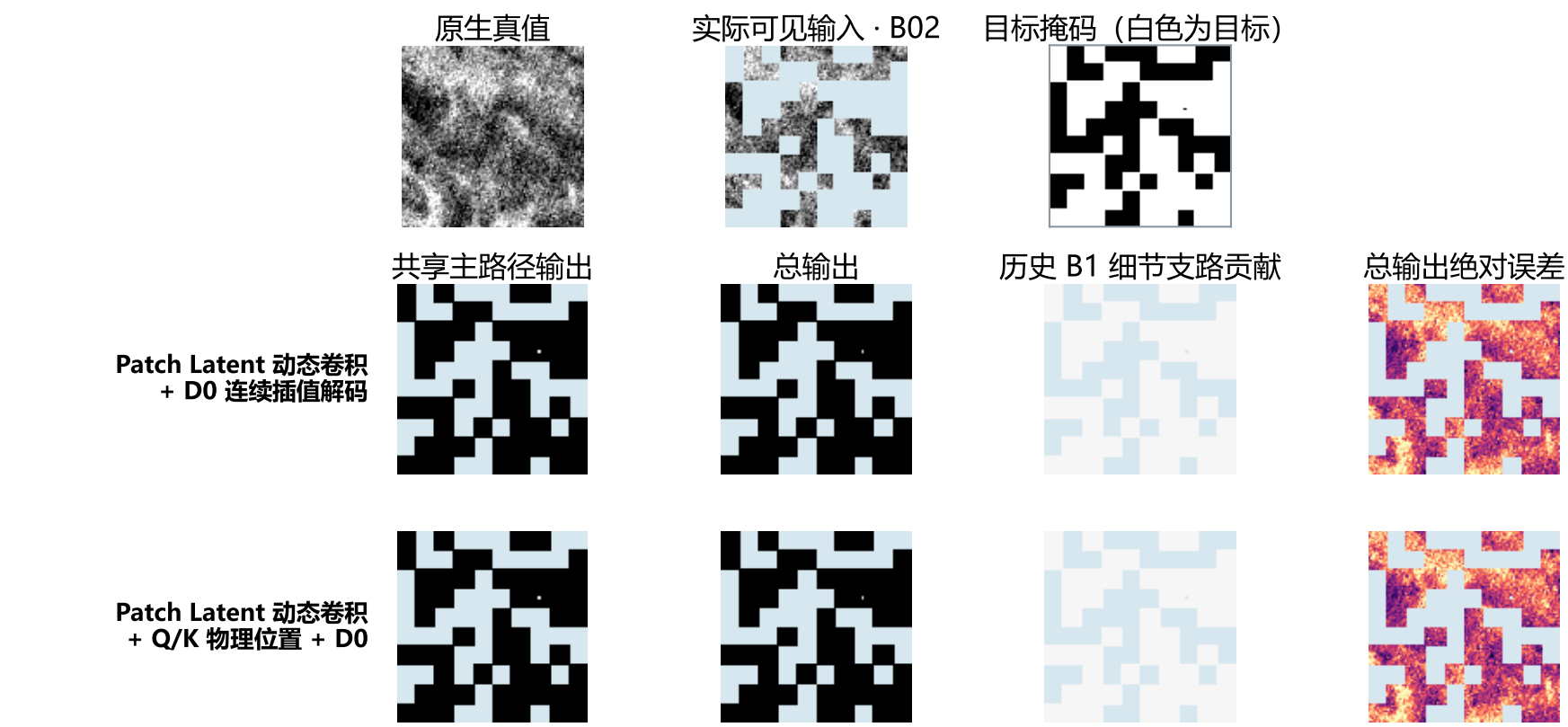


该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S6 • A • S2 • B02 • 第 0 步

obs_5929ceec240fa570c7a1a784

六种模型架构 · 同观测、同目标掩码、同真值色标

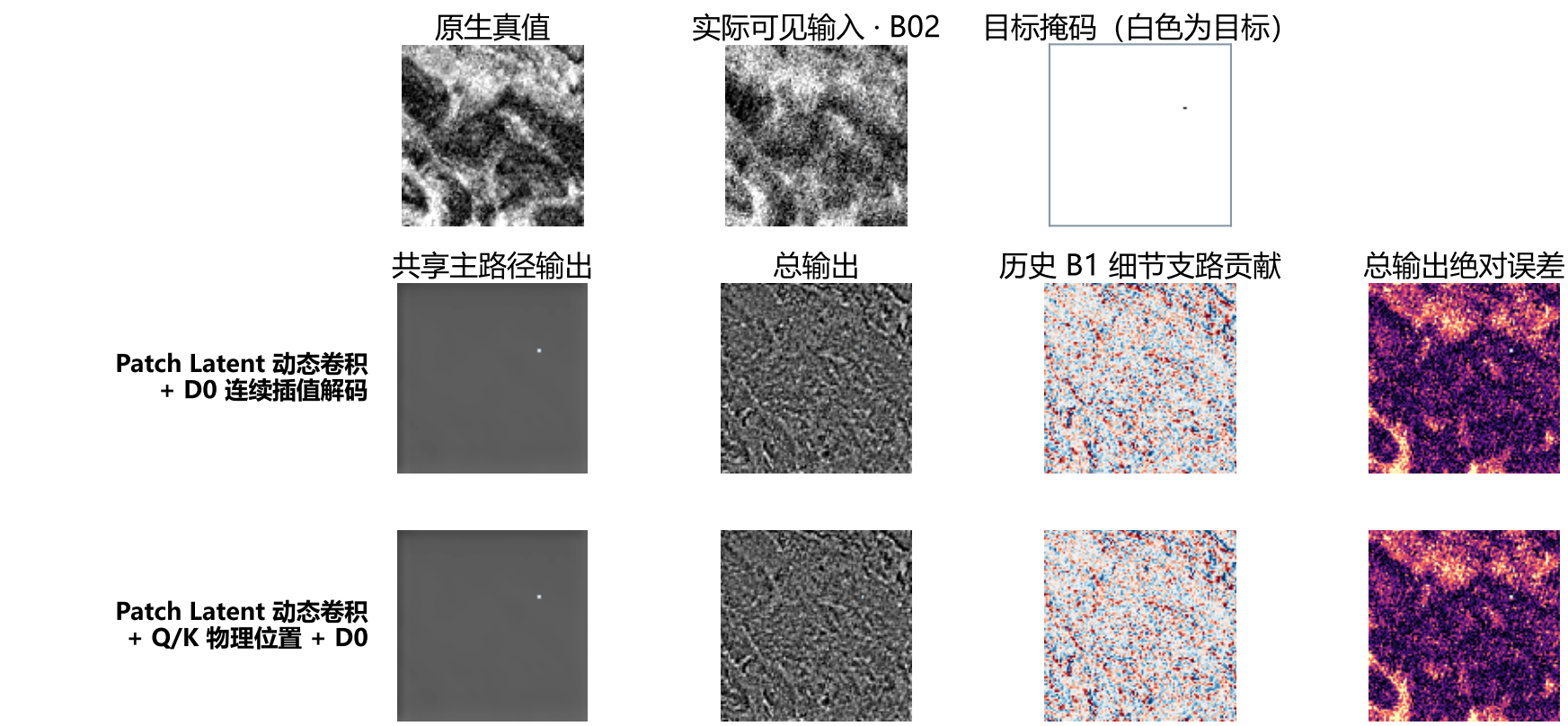


该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S6 • B1 • S2 • B03 • 第 0 步

obs_5929ceec240fa570c7a1a784

六种模型架构 · 同观测、同目标掩码、同真值色标

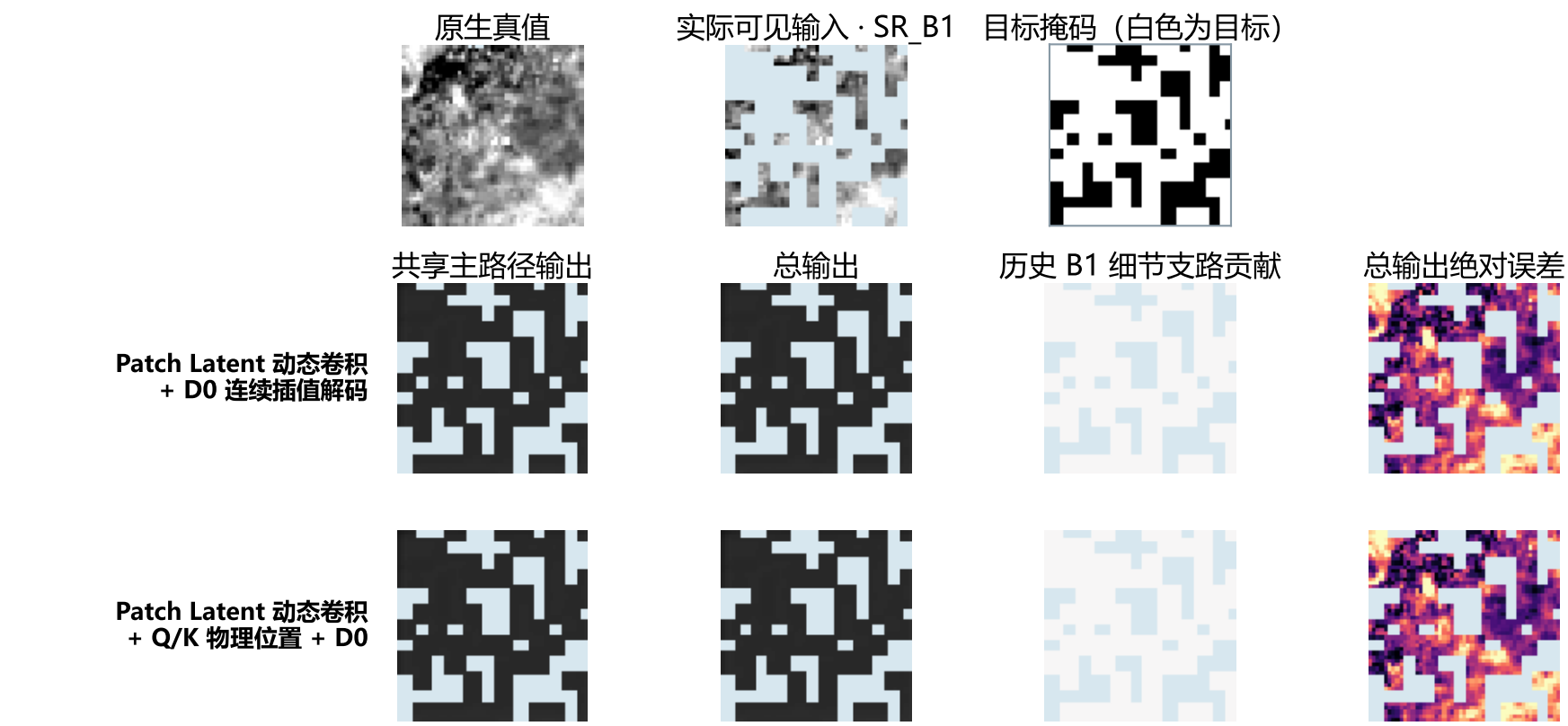


该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S6 • A • L8 • SR_B1 • 第 0 步

obs_64d10dd286169cf55d8668ac

六种模型架构 · 同观测、同目标掩码、同真值色标

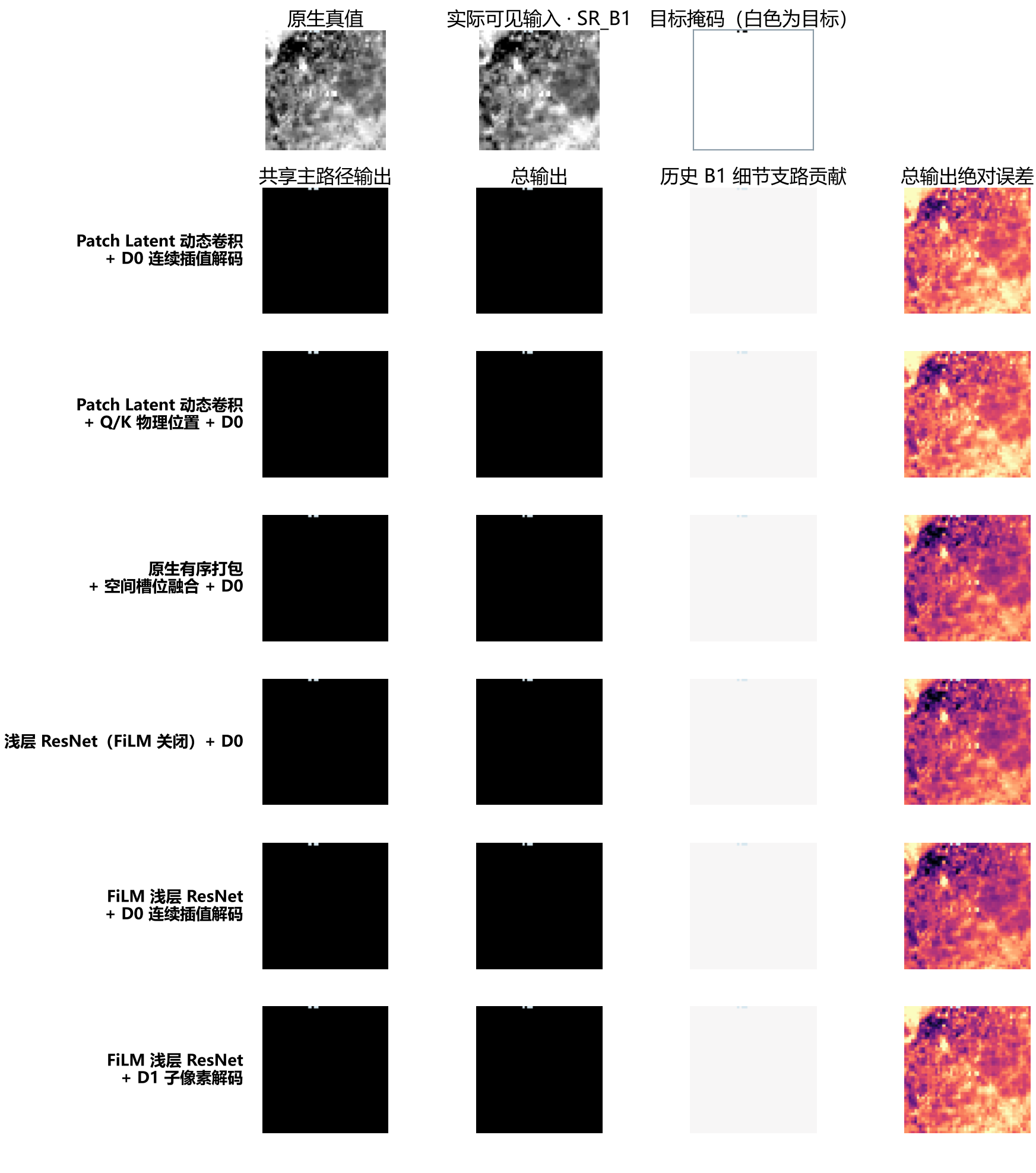


该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S6 • B1 • L8 • SR_B2 • 第 0 步

obs_64d10dd286169cf55d8668ac

六种模型架构 · 同观测、同目标掩码、同真值色标

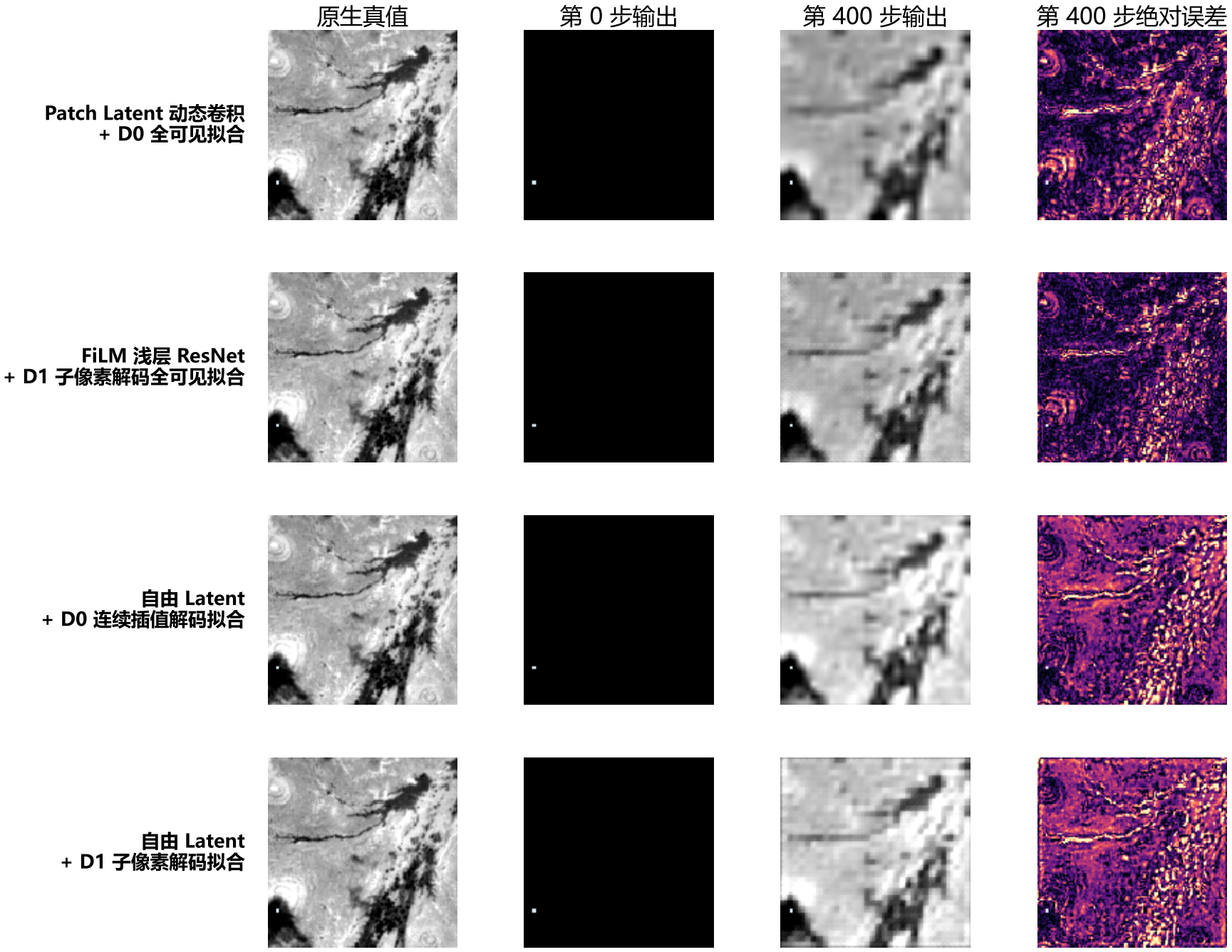


该图选取该观测目标像元数最多的波段。浅蓝色为无预测或无效像元。误差与支路贡献各用共享色标。

S5 · S2 · B02 · TRAIN 拟合样本

obs_53a1c1f77f42ff440b872b9e

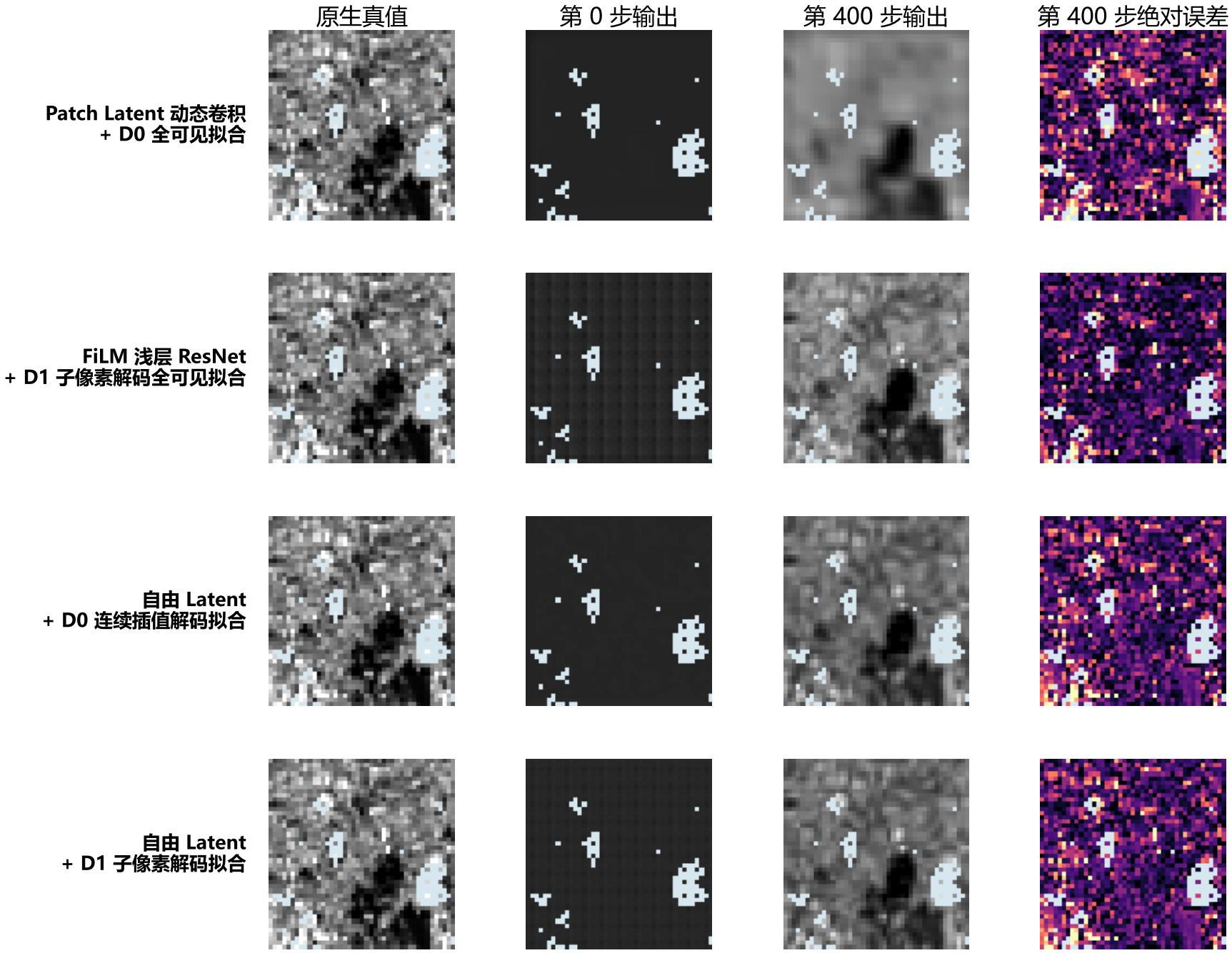
统一展示第 400 步 · 历史 B1 细节支路关闭



S5 • L8 • SR_B2 • TRAIN 拟合样本

obs_3f17b1a68076c7f1176b0abe_L8

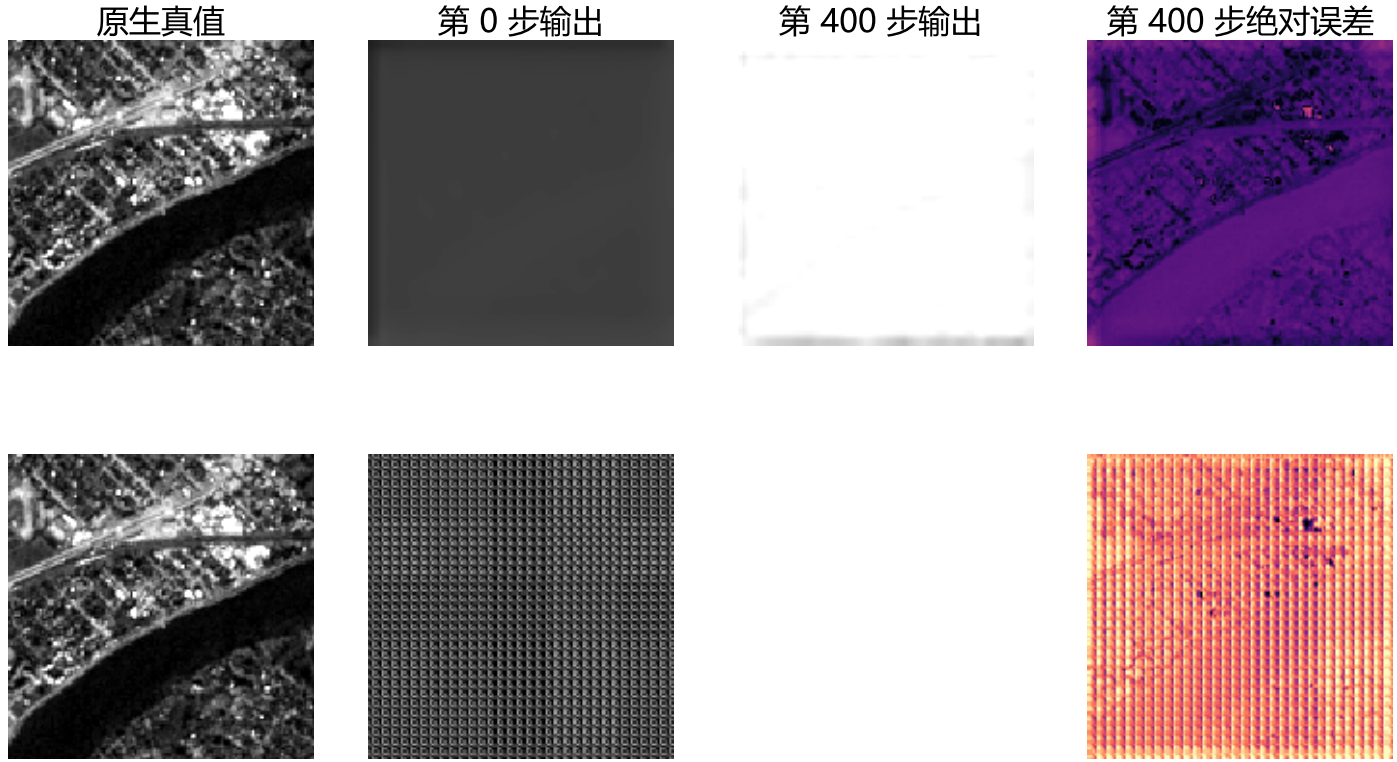
统一展示第 400 步 · 历史 B1 细节支路关闭



S5 · S2 · B02 · TRAIN 只读样本

obs_801233e3b7b1ebe72b0a2611

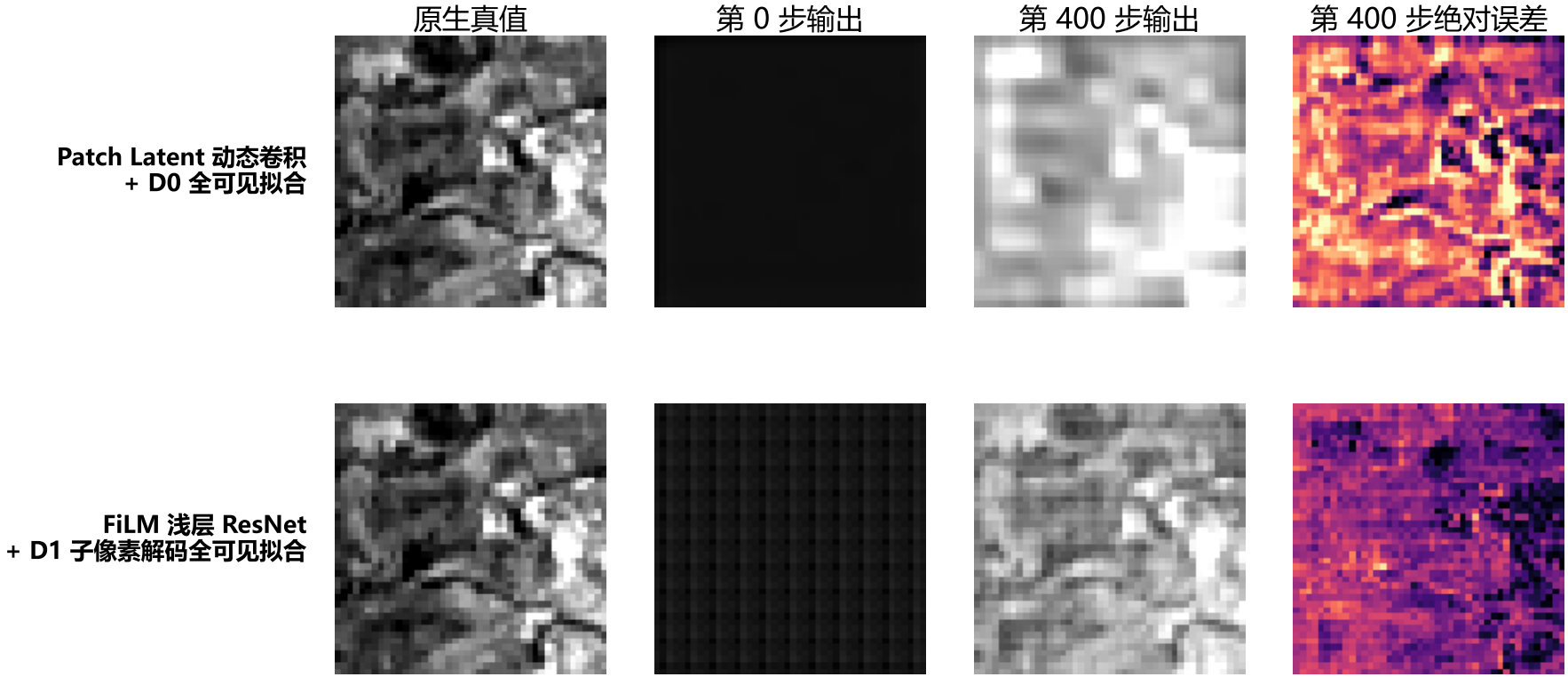
统一展示第 400 步 · 历史 B1 细节支路关闭



S5 · L8 · SR_B2 · TRAIN 只读样本

obs_b67199db7864a8f00243f660_L8

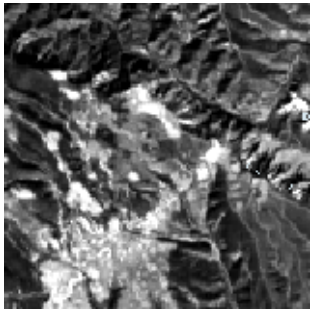
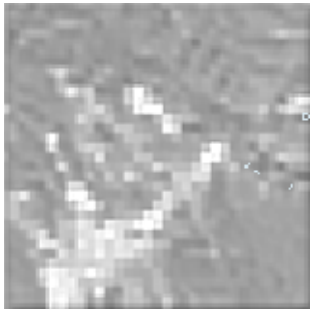
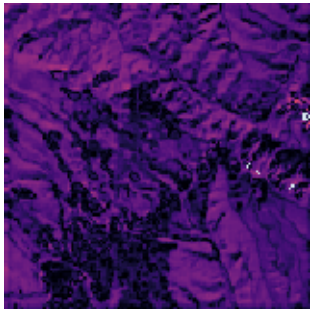
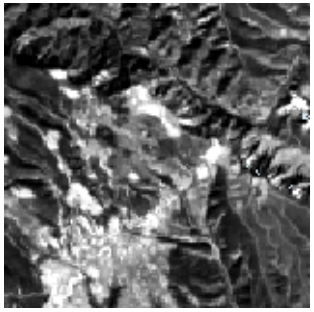
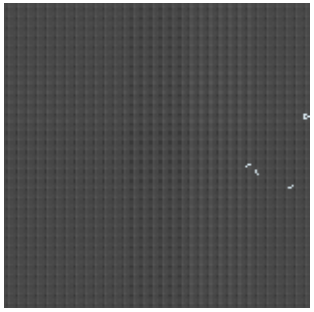
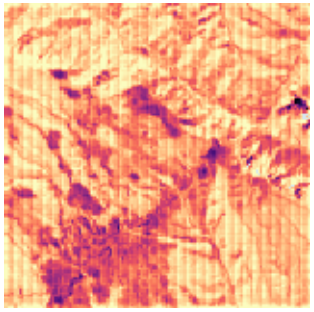
统一展示第 400 步 · 历史 B1 细节支路关闭



S5 · L9 · SR_B2 · TRAIN 只读样本

obs_176447f96f63c87282265153

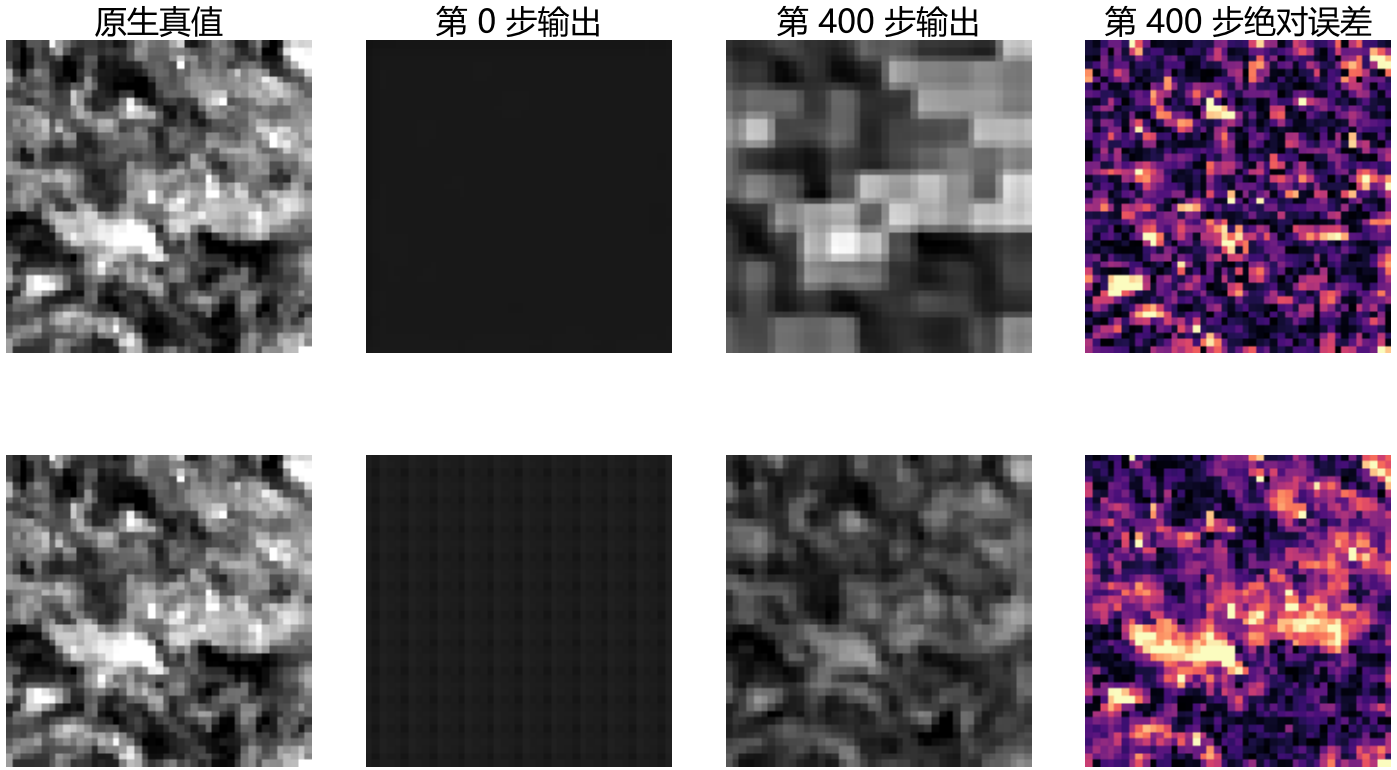
统一展示第 400 步 · 历史 B1 细节支路关闭

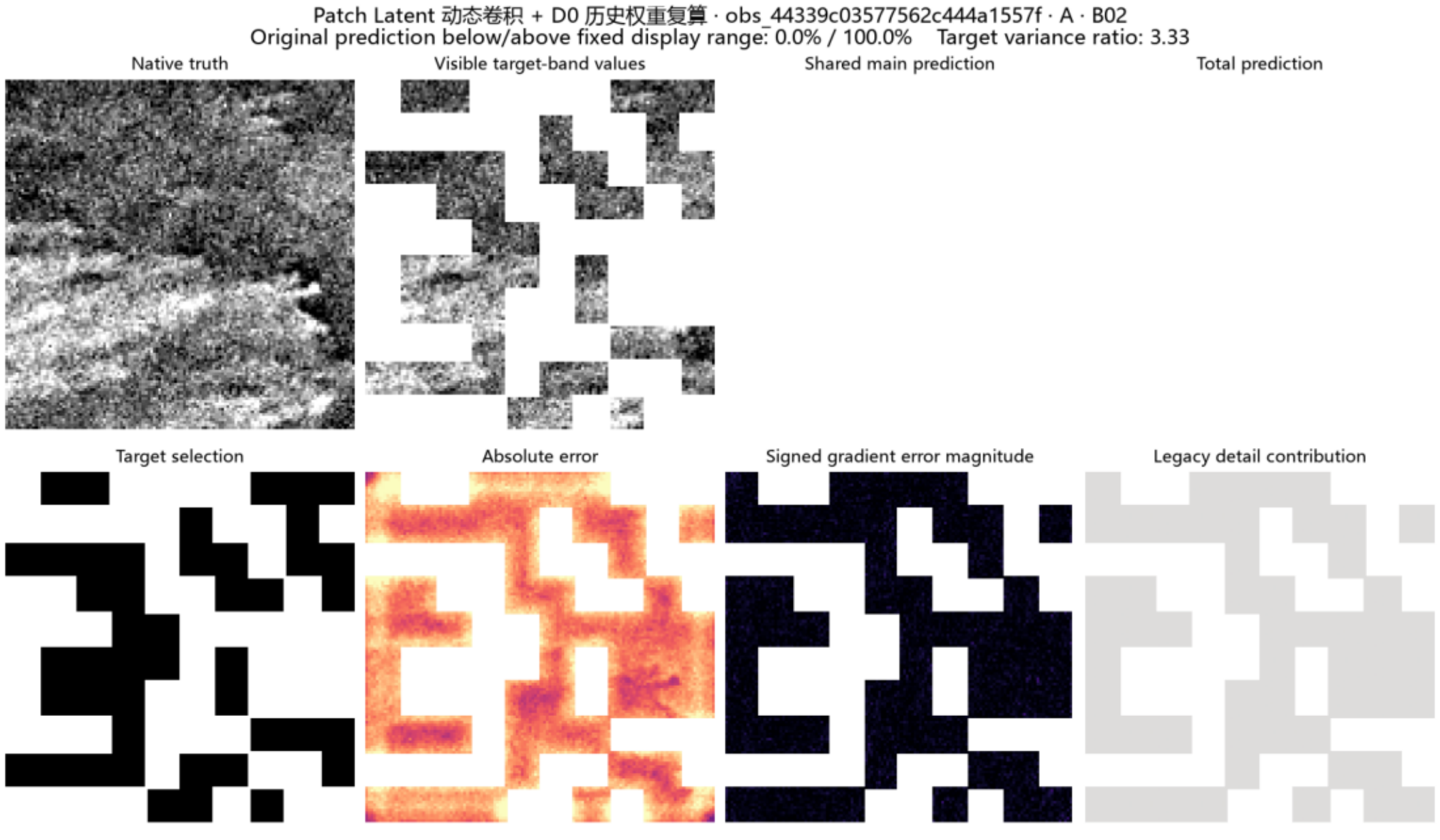
	原生真值	第 0 步输出	第 400 步输出	第 400 步绝对误差
Patch Latent 动态卷积 + D0 全可见拟合				
FiLM 浅层 ResNet + D1 子像素解码全可见拟合				

S5 · EnMAP · B001 · TRAIN 只读样本

enmap_native_chimanimani_s2_1060

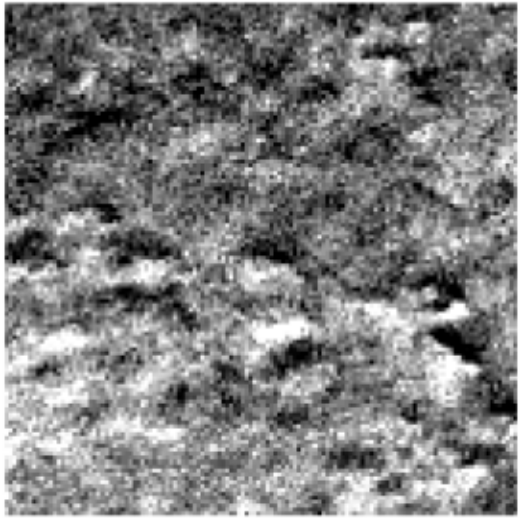
统一展示第 400 步 · 历史 B1 细节支路关闭




Patch Latent 动态卷积 + D0 历史权重复算 · obs_44339c03577562c444a1557f · A · B02
Original prediction below/above fixed display range: 0.0% / 100.0%
Target variance ratio: 3.33
Native truth
Visible target-band values
Shared main prediction
Total prediction
Target selection
Absolute error
Signed gradient error magnitude
Legacy detail contribution

Patch Latent 动态卷积 + D0 历史权重复算 · obs_44339c03577562c444a1557f · B1 · B04
Original prediction below/above fixed display range: 100.0% / 0.0% Target variance ratio: 2.72

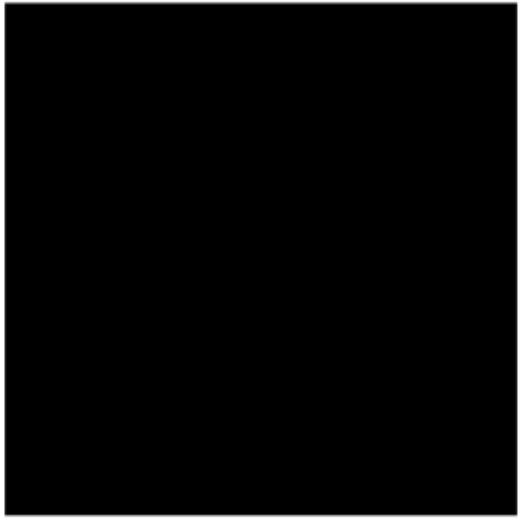
Native truth



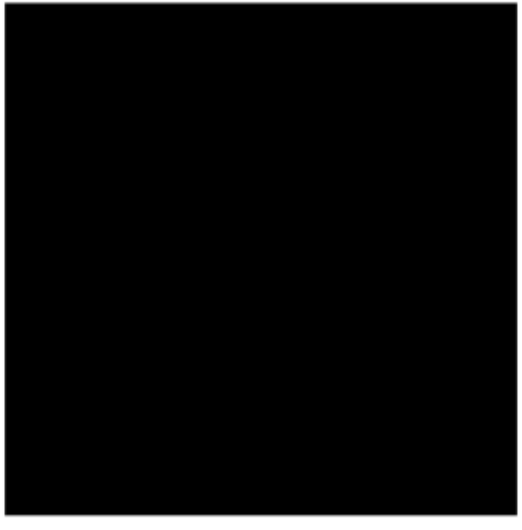
Visible reference B02



Shared main prediction

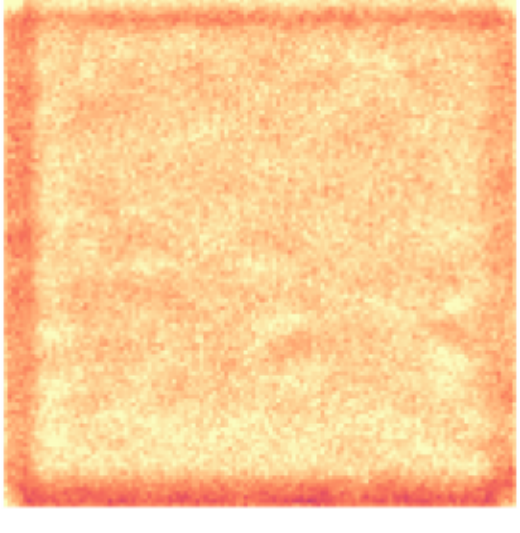


Total prediction

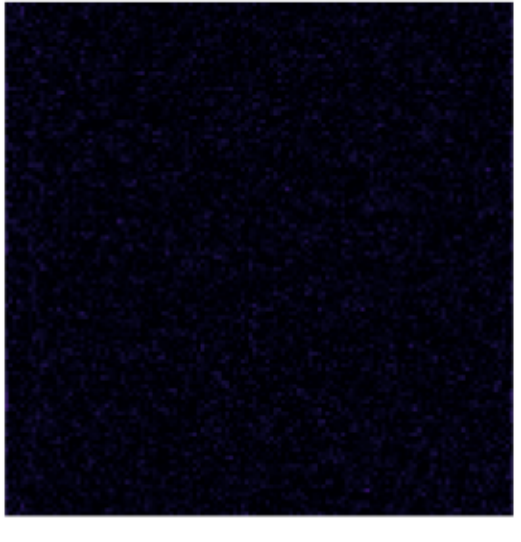


Target selection

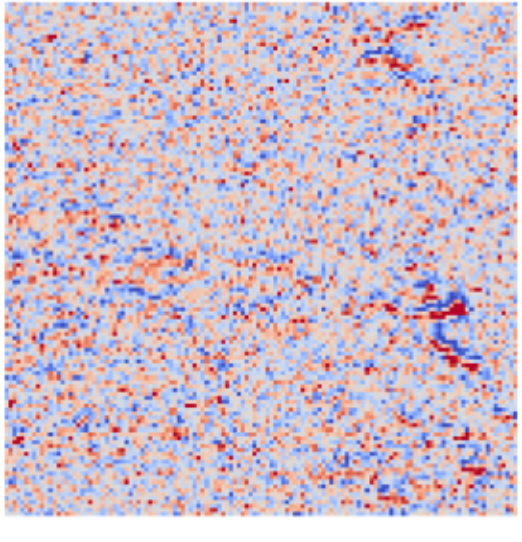
Absolute error

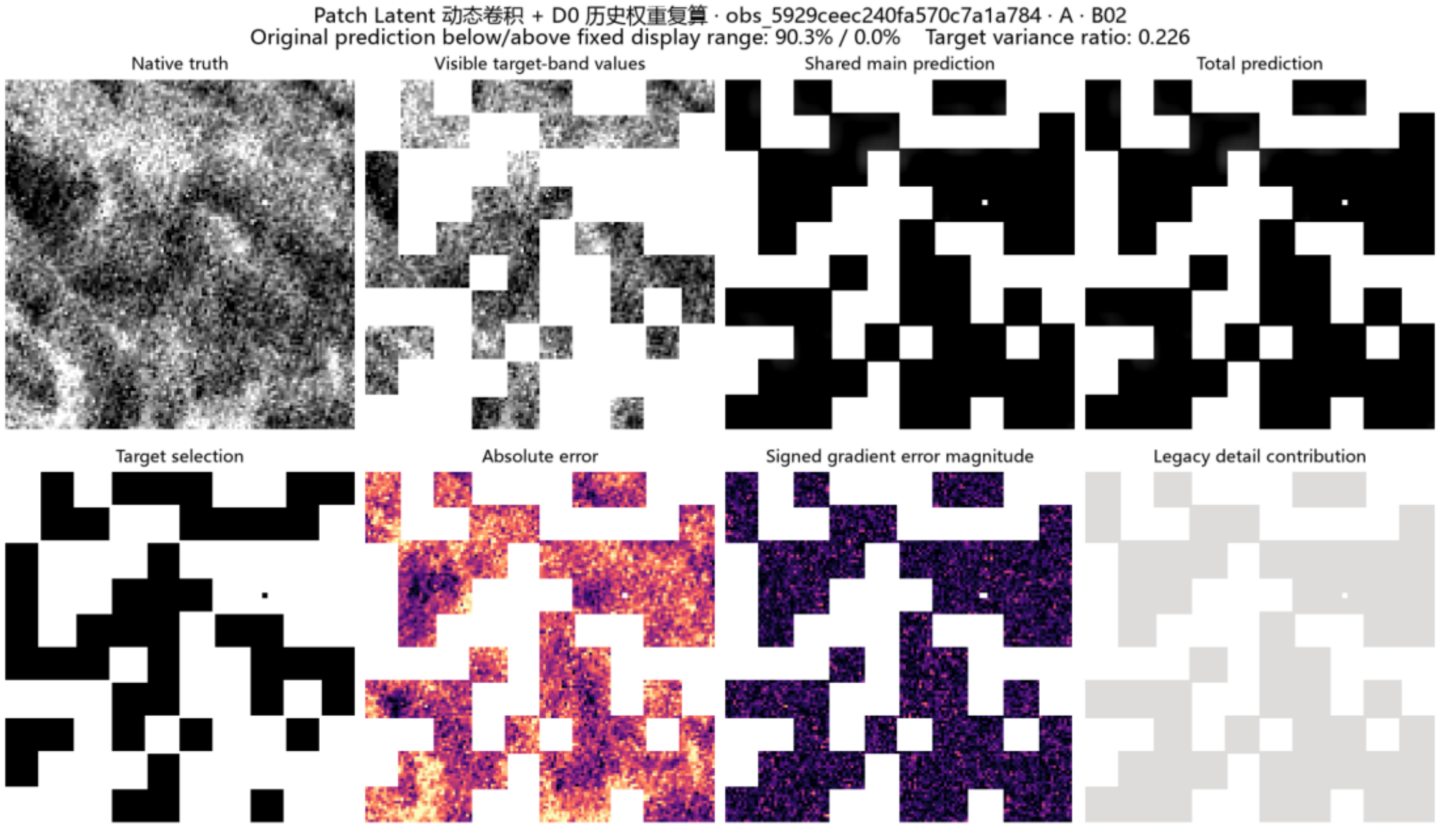


Signed gradient error magnitude



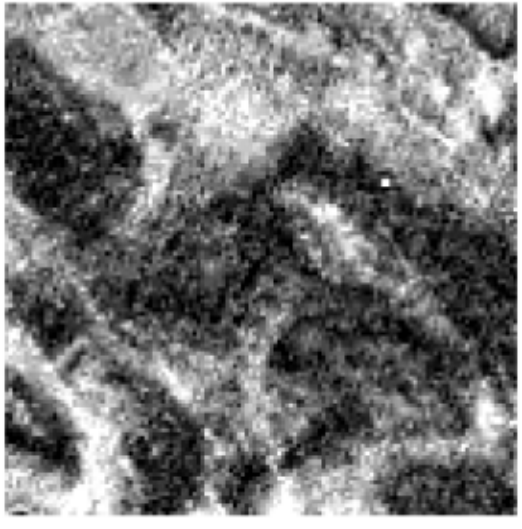
Legacy detail contribution



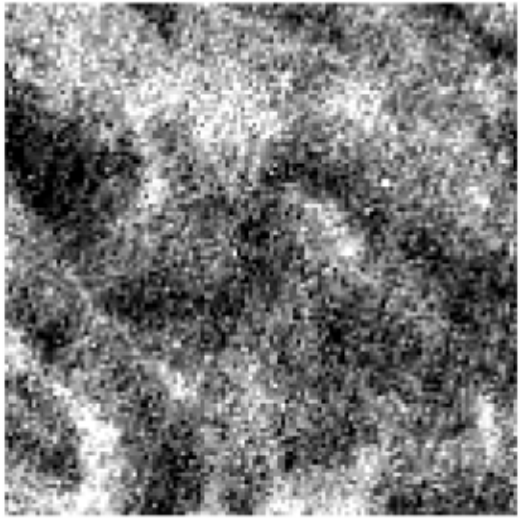

Patch Latent 动态卷积 + D0 历史权重复算 · obs_5929ceec240fa570c7a1a784 · A · B02
Original prediction below/above fixed display range: 90.3% / 0.0% Target variance ratio: 0.226
Native truth
Visible target-band values
Shared main prediction
Total prediction
Target selection
Absolute error
Signed gradient error magnitude
Legacy detail contribution

Patch Latent 动态卷积 + D0 历史权重复算 · obs_5929ceec240fa570c7a1a784 · B1 · B03
Original prediction below/above fixed display range: 0.0% / 40.6% Target variance ratio: 2.39

Native truth



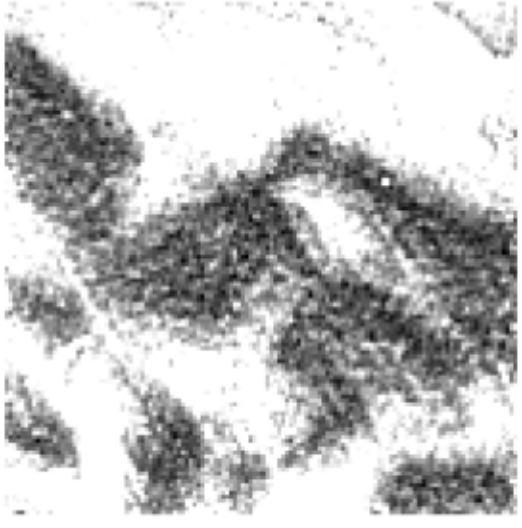
Visible reference B02



Shared main prediction

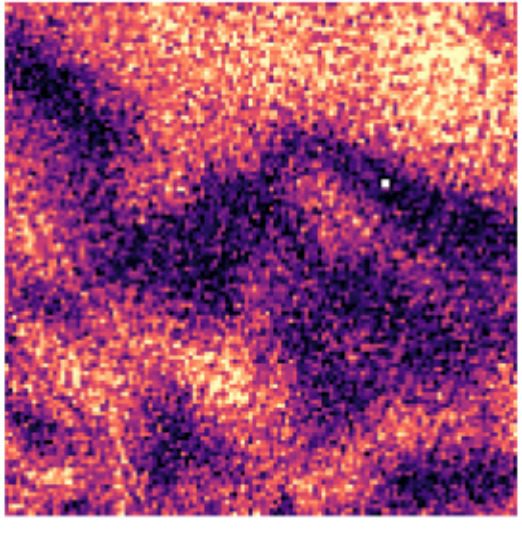


Total prediction

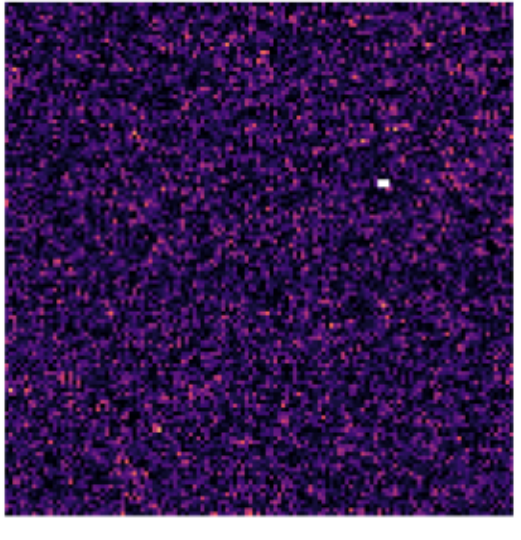


Target selection

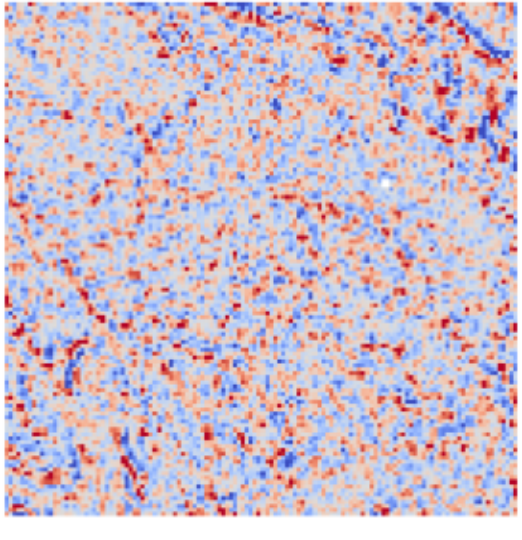
Absolute error

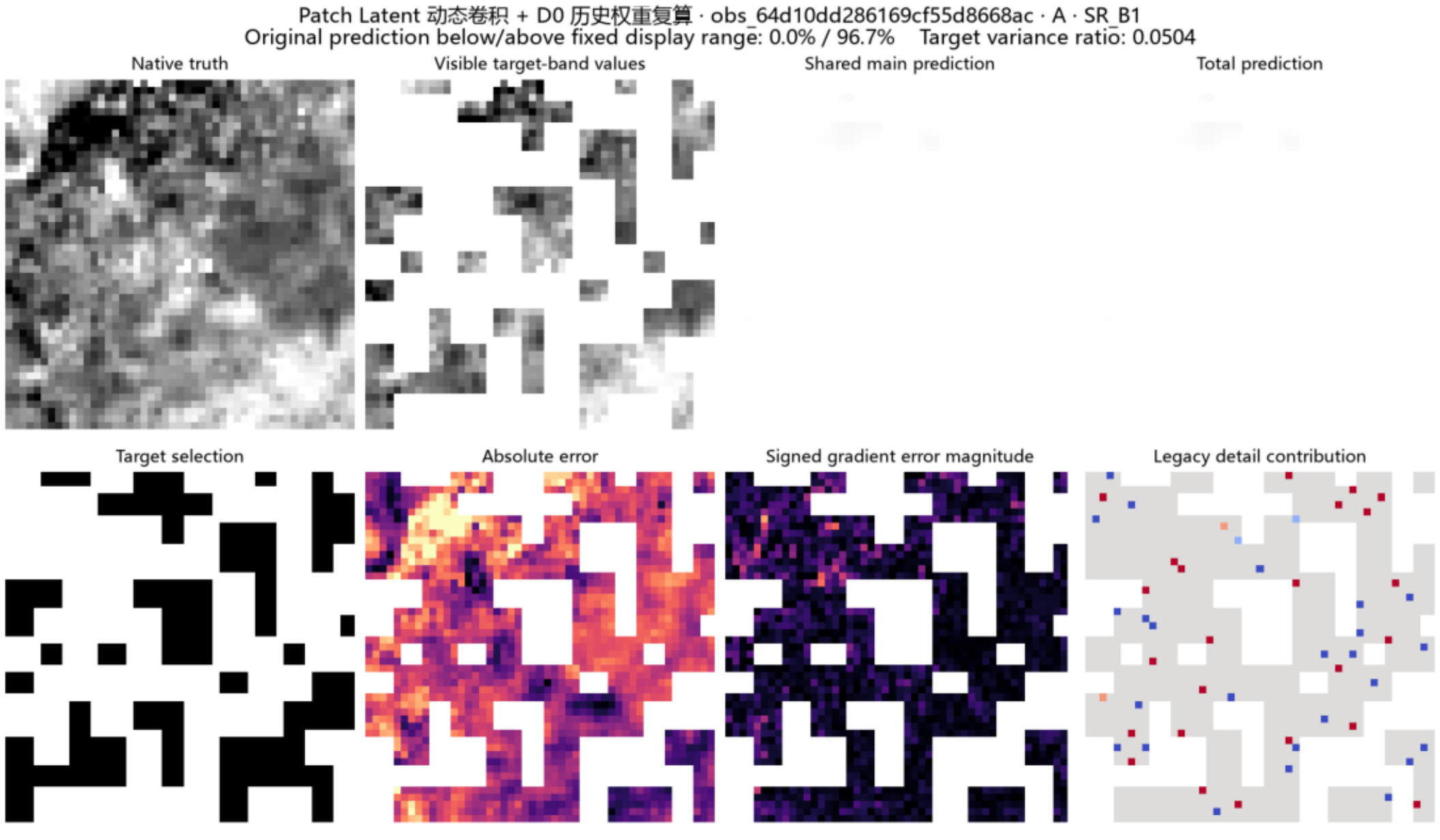


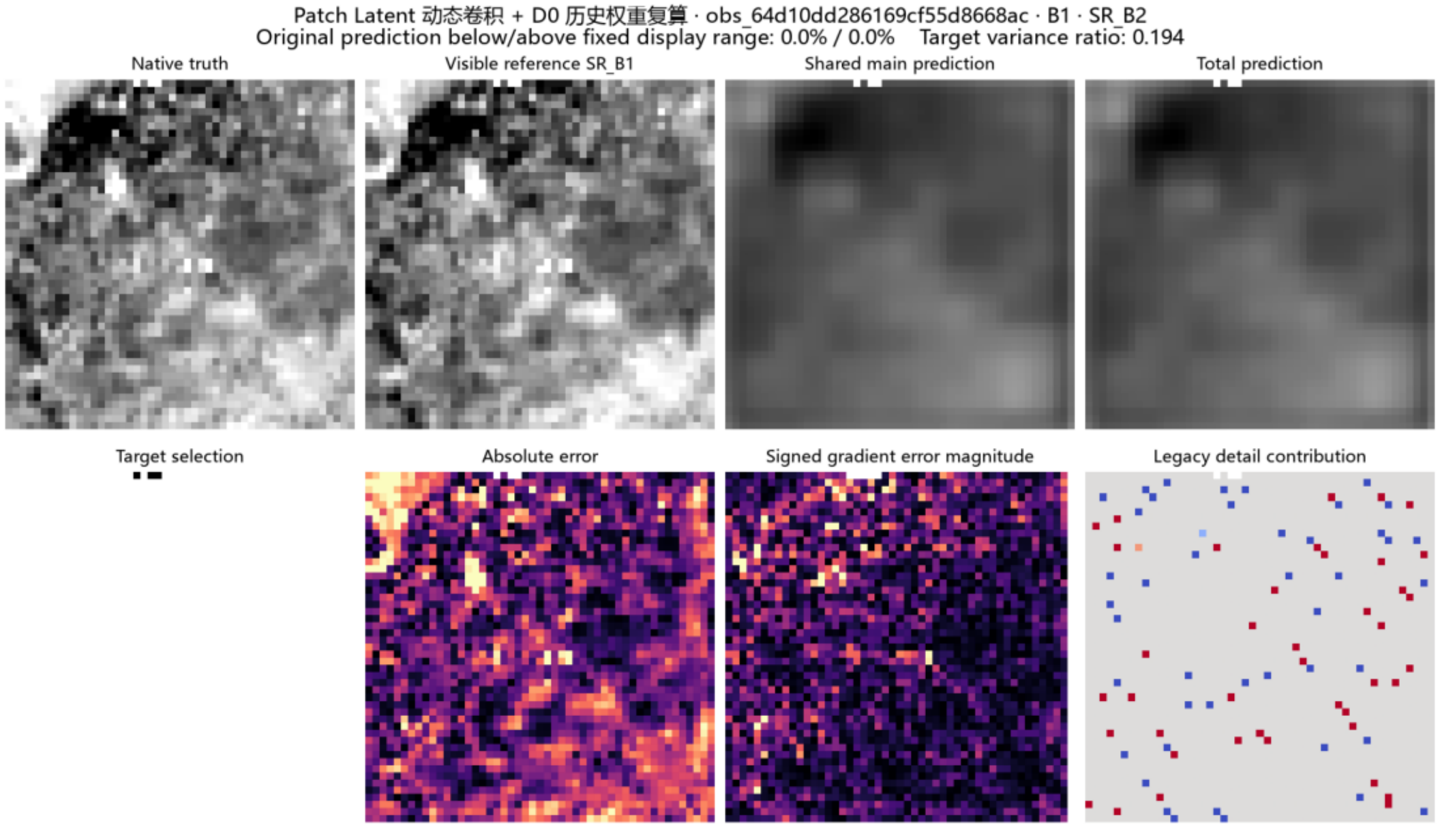
Signed gradient error magnitude

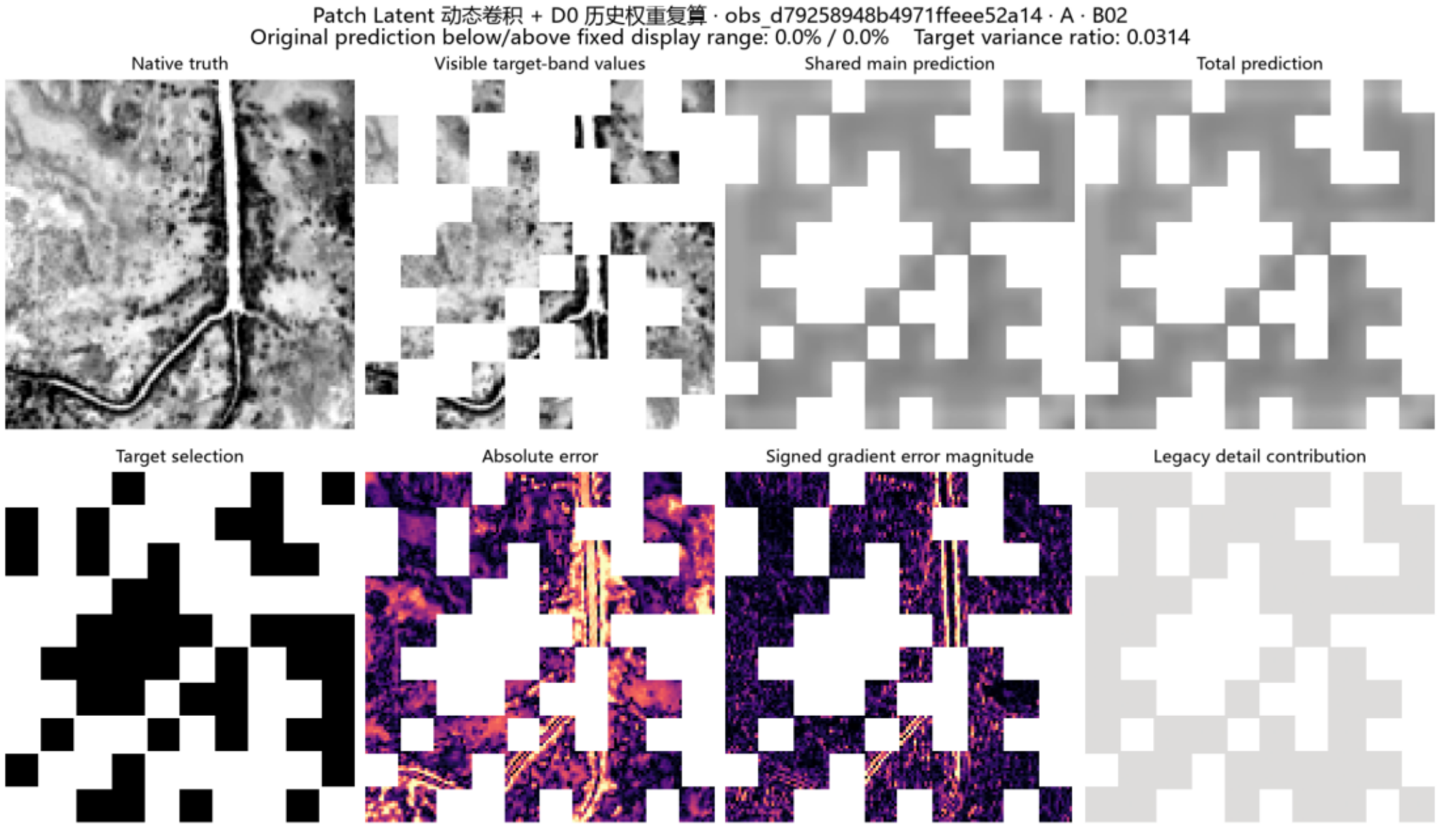


Legacy detail contribution



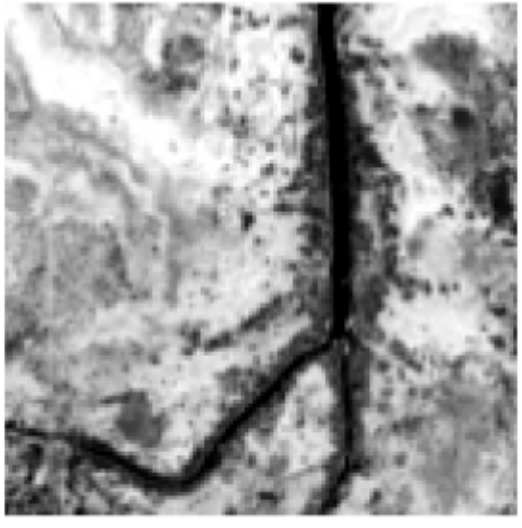

Patch Latent 动态卷积 + D0 历史权重复算 · obs_64d10dd286169cf55d8668ac · A · SR_B1
Original prediction below/above fixed display range: 0.0% / 96.7%
Target variance ratio: 0.0504
Native truth
Visible target-band values
Shared main prediction
Total prediction
Target selection
Absolute error
Signed gradient error magnitude
Legacy detail contribution


Patch Latent 动态卷积 + D0 历史权重复算 · obs_64d10dd286169cf55d8668ac · B1 · SR_B2
Original prediction below/above fixed display range: 0.0% / 0.0%
Target variance ratio: 0.194
Native truth
Visible reference SR_B1
Shared main prediction
Total prediction
Target selection
Absolute error
Signed gradient error magnitude
Legacy detail contribution

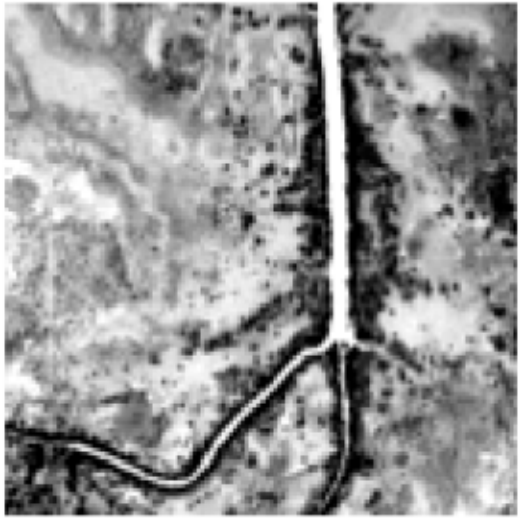

Patch Latent 动态卷积 + D0 历史权重复算 · obs_d79258948b4971ffeee52a14 · A · B02
Original prediction below/above fixed display range: 0.0% / 0.0%
Target variance ratio: 0.0314
Native truth
Visible target-band values
Shared main prediction
Total prediction
Target selection
Absolute error
Signed gradient error magnitude
Legacy detail contribution

Patch Latent 动态卷积 + D0 历史权重复算 · obs_d79258948b4971ffeee52a14 · B1 · B08
Original prediction below/above fixed display range: 0.0% / 16.7% Target variance ratio: 0.717

Native truth



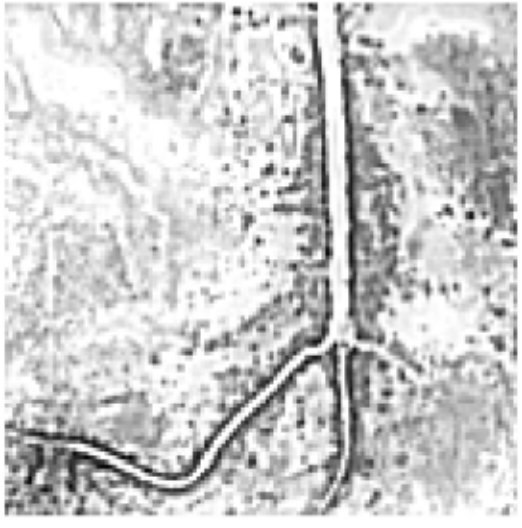
Visible reference B02



Shared main prediction

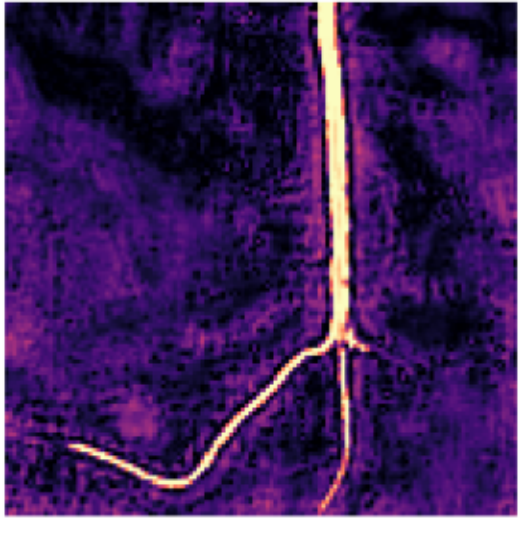


Total prediction

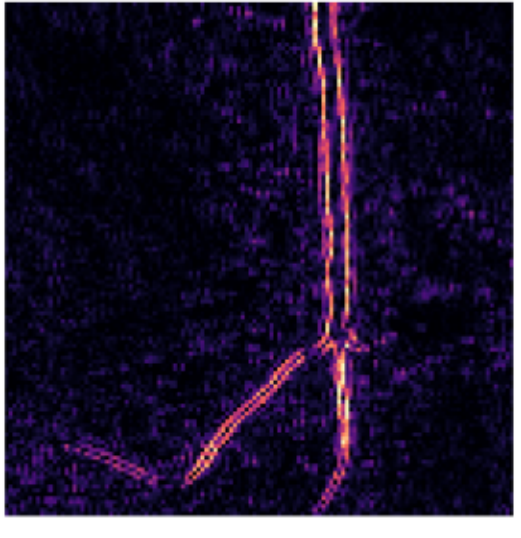


Target selection

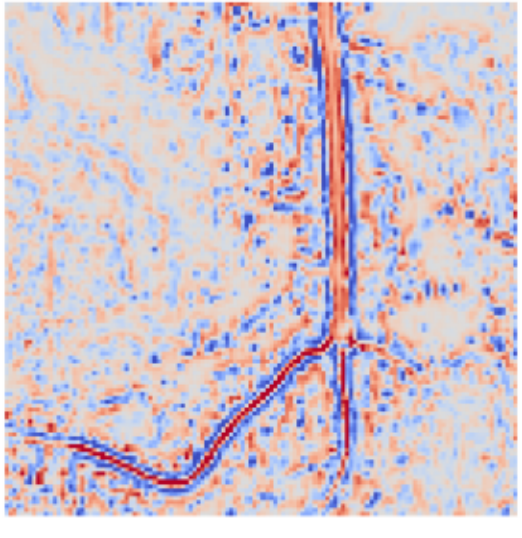
Absolute error

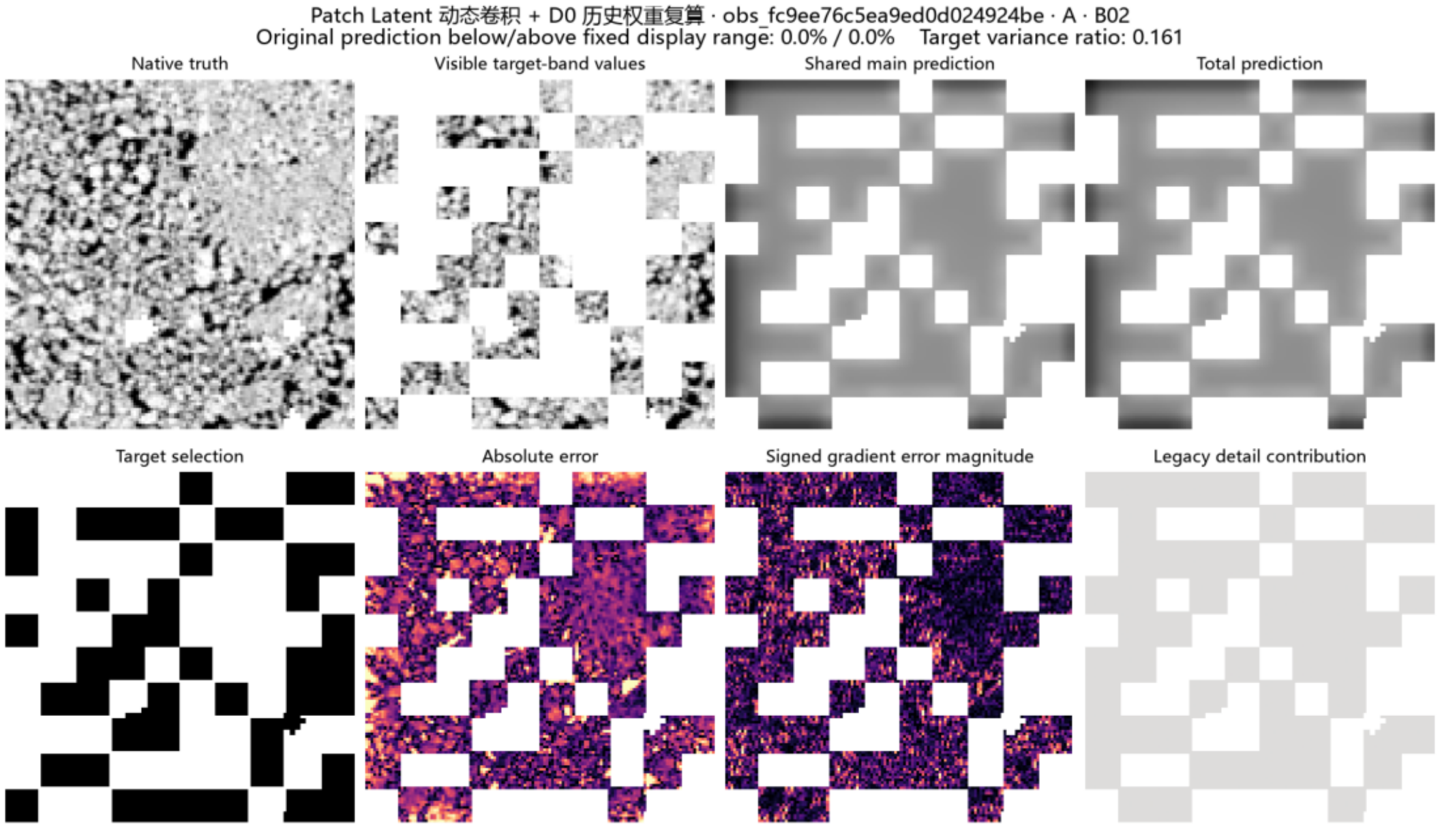


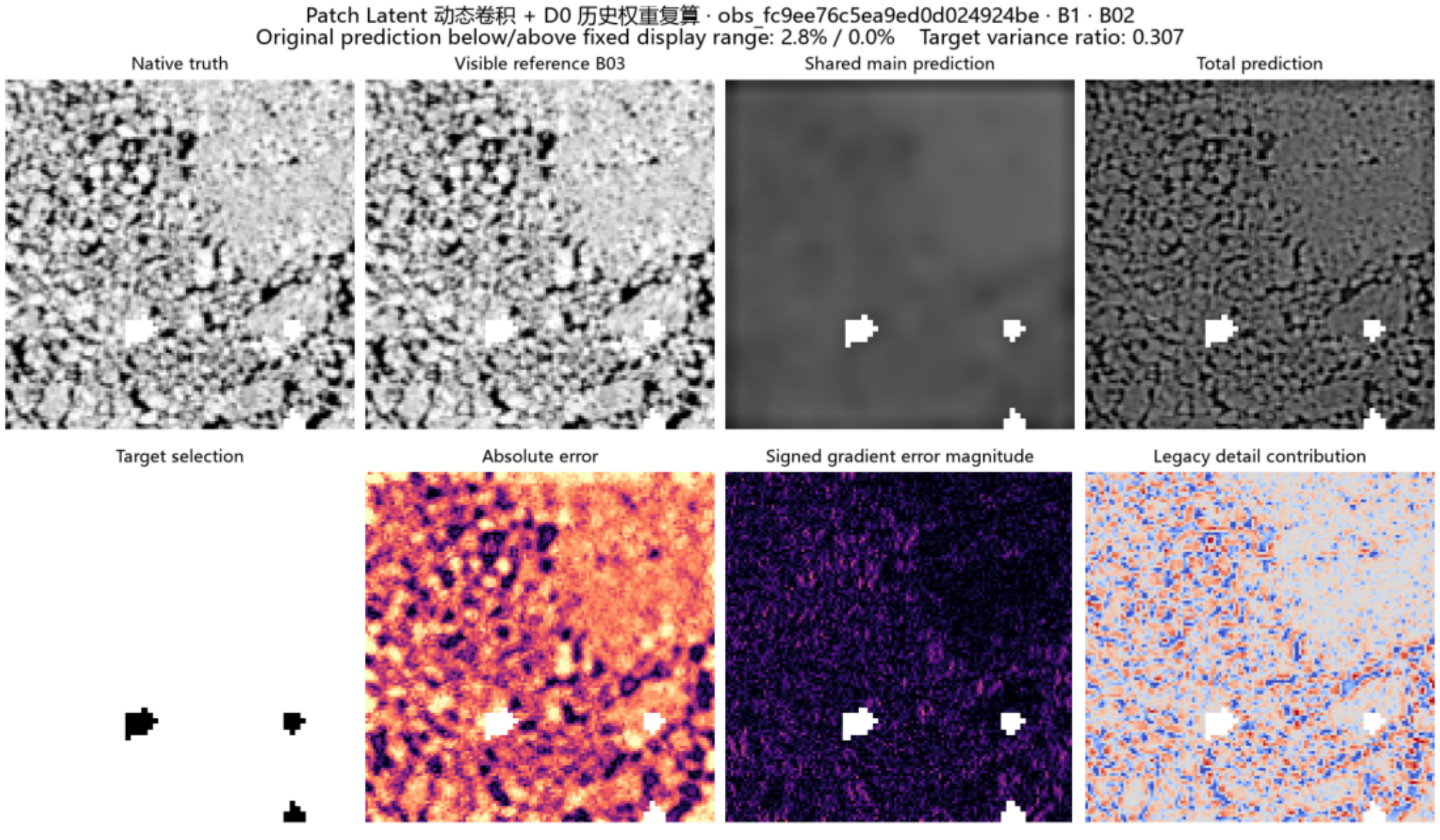
Signed gradient error magnitude



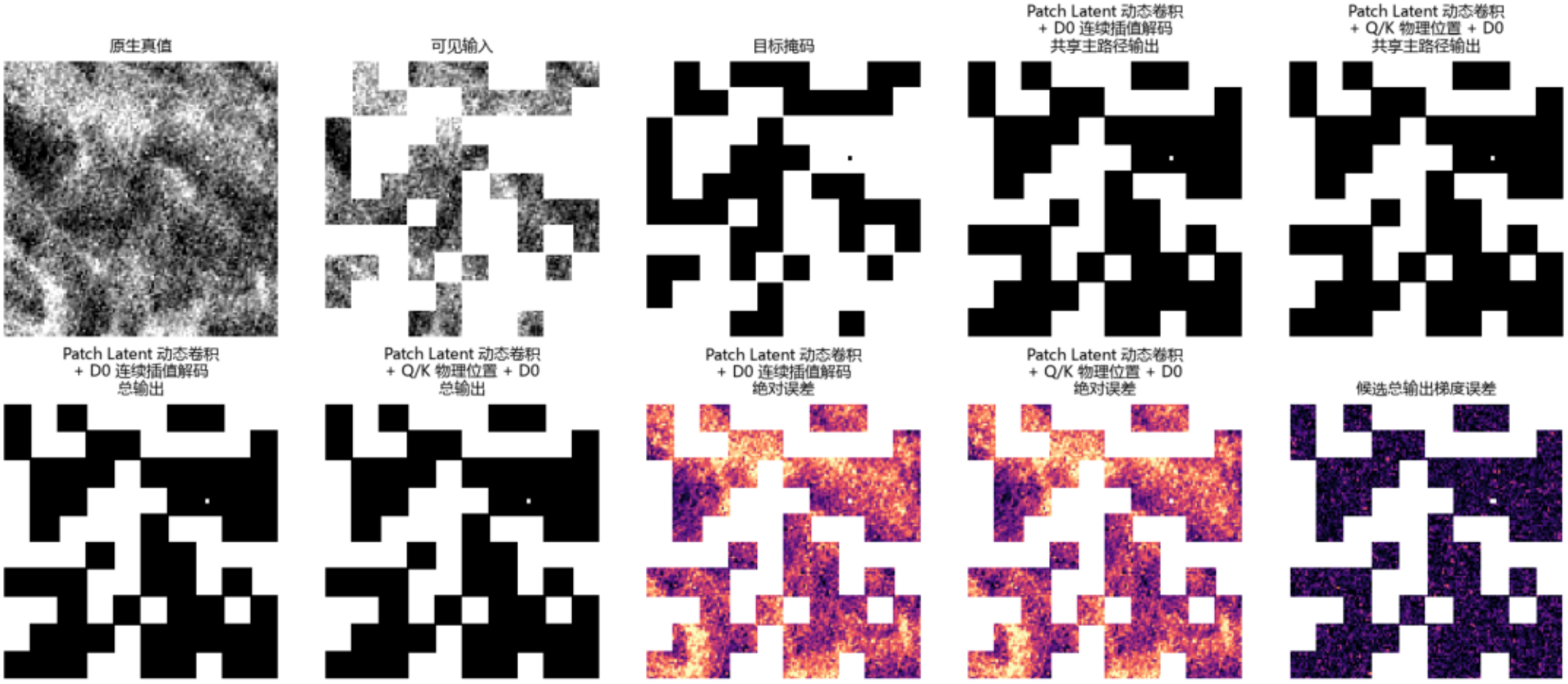
Legacy detail contribution




Patch Latent 动态卷积 + D0 历史权重复算 · obs_fc9ee76c5ea9ed0d024924be · A · B02
Original prediction below/above fixed display range: 0.0% / 0.0% Target variance ratio: 0.161
Native truth
Visible target-band values
Shared main prediction
Total prediction
Target selection
Absolute error
Signed gradient error magnitude
Legacy detail contribution

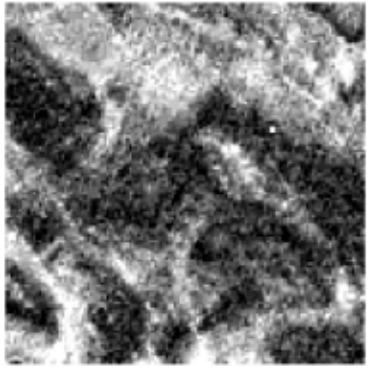

Patch Latent 动态卷积 + D0 历史权重复算 · obs_fc9ee76c5ea9ed0d024924be · B1 · B02
Original prediction below/above fixed display range: 2.8% / 0.0% Target variance ratio: 0.307
Native truth
Visible reference B03
Shared main prediction
Total prediction
Target selection
Absolute error
Signed gradient error magnitude
Legacy detail contribution

V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_A_B02

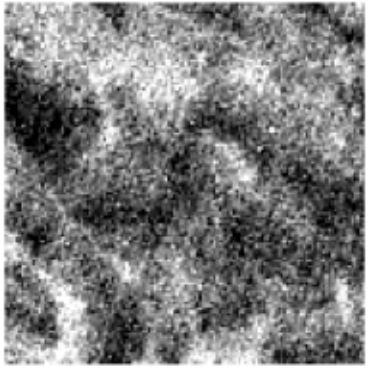


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_B1_B03

原生真值



可见 B02



目标掩码

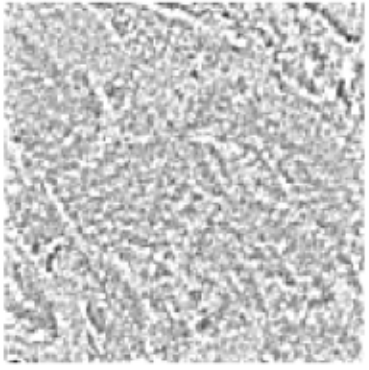
Patch Latent 动态卷积
+ D0 连续插值解码
共享主路径输出



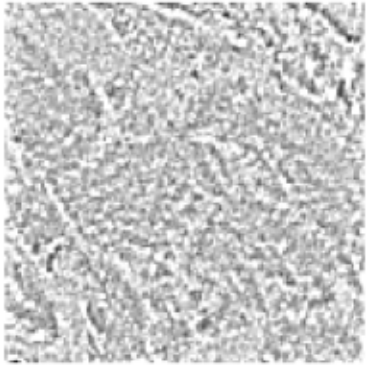
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
共享主路径输出



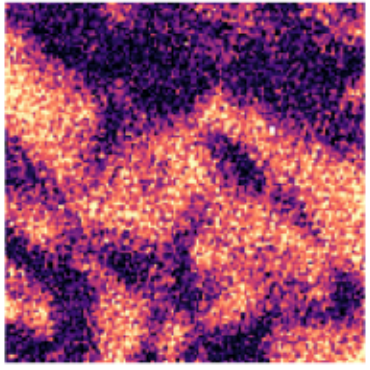
Patch Latent 动态卷积
+ D0 连续插值解码
叠加历史支路后的总输出



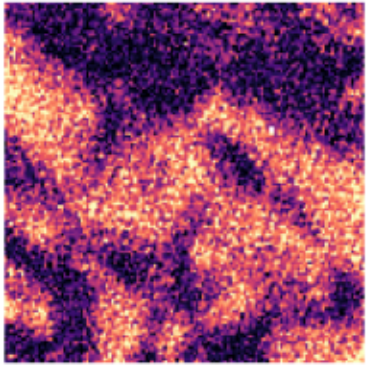
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
叠加历史支路后的总输出



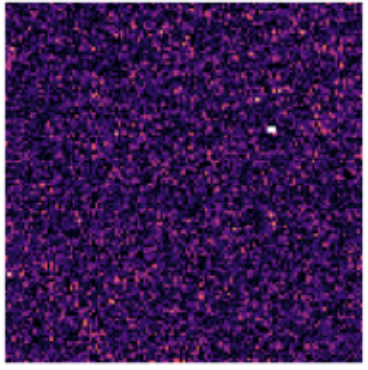
Patch Latent 动态卷积
+ D0 连续插值解码
绝对误差



Patch Latent 动态卷积
+ Q/K 物理位置 + D0
绝对误差

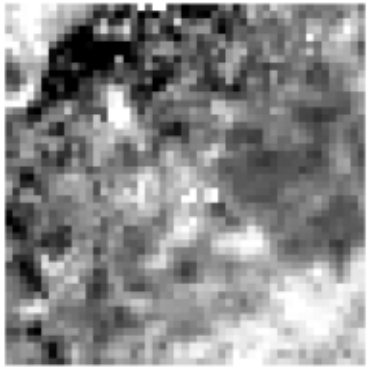


候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_A_SR_B1

原生真值



可见输入



目标掩码



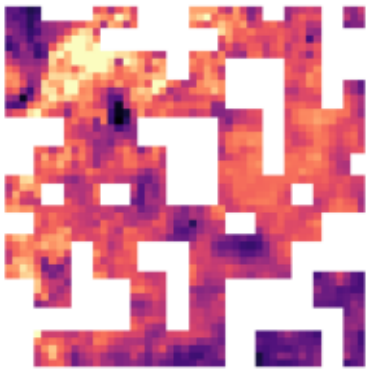
Patch Latent 动态卷积
+ D0 连续插值解码
共享主路径输出

Patch Latent 动态卷积
+ Q/K 物理位置 + D0
共享主路径输出

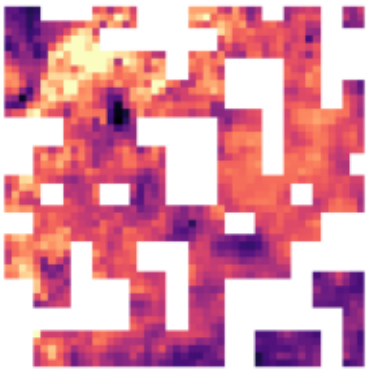
Patch Latent 动态卷积
+ D0 连续插值解码
总输出

Patch Latent 动态卷积
+ Q/K 物理位置 + D0
总输出

Patch Latent 动态卷积
+ D0 连续插值解码
绝对误差



Patch Latent 动态卷积
+ Q/K 物理位置 + D0
绝对误差



候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_B1_SR_B2

原生真值



可见 SR_B1



目标掩码



Patch Latent 动态卷积
+ D0 连续插值解码
共享主路径输出



Patch Latent 动态卷积
+ Q/K 物理位置 + D0
共享主路径输出



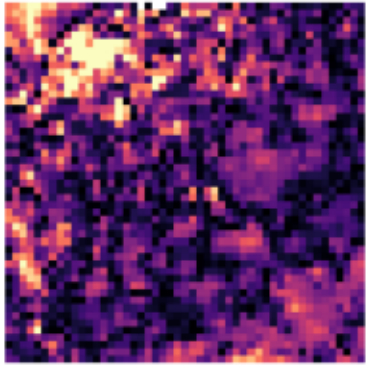
Patch Latent 动态卷积
+ D0 连续插值解码
叠加历史支路后的总输出



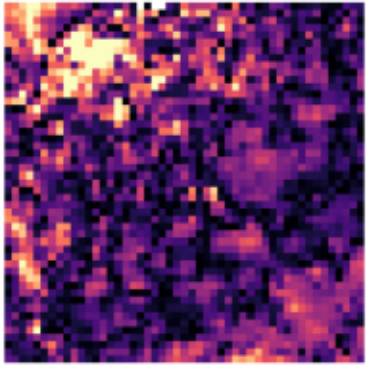
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
叠加历史支路后的总输出



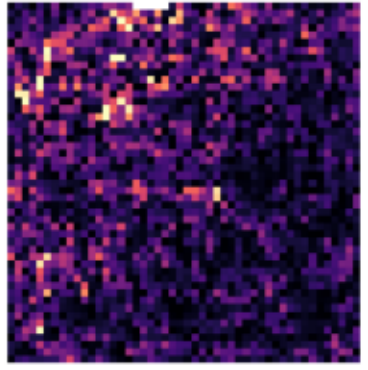
Patch Latent 动态卷积
+ D0 连续插值解码
绝对误差



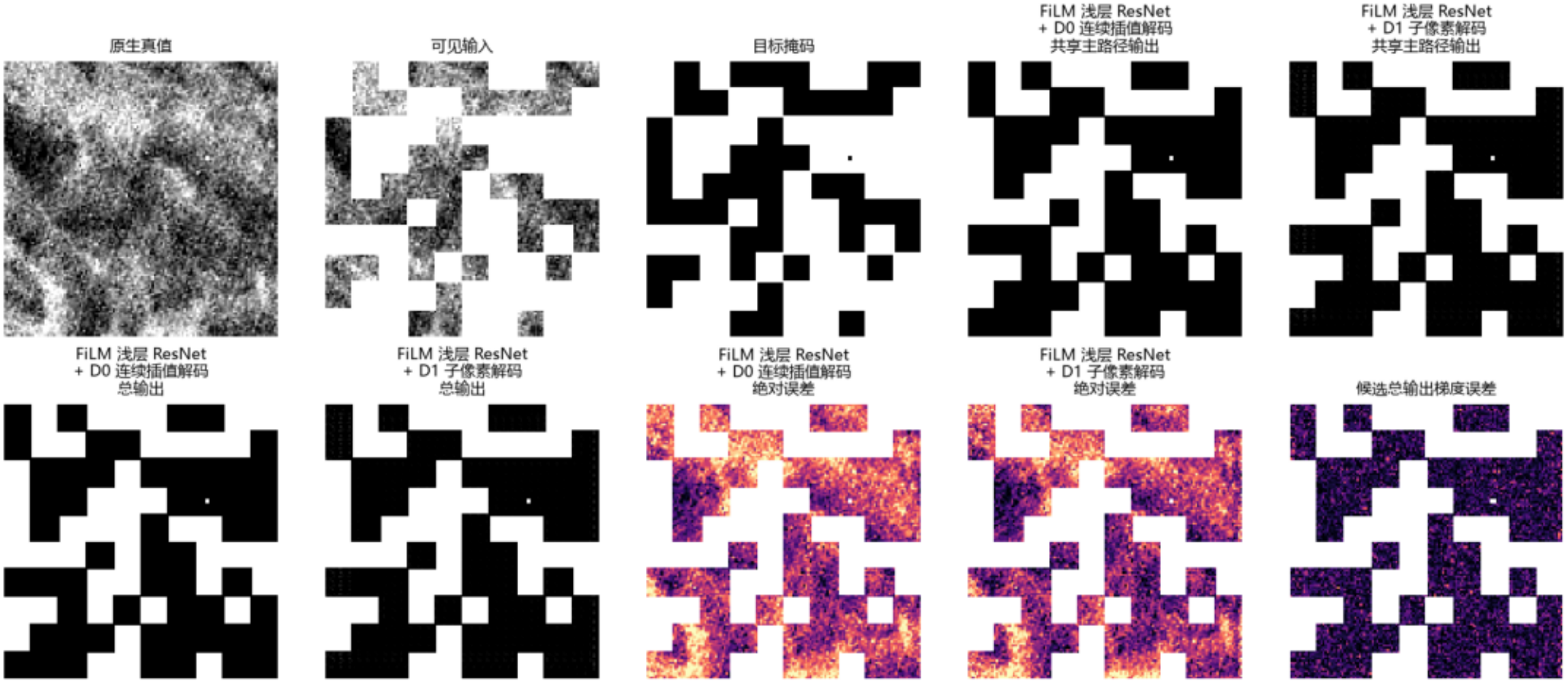
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
绝对误差



候选总输出梯度误差

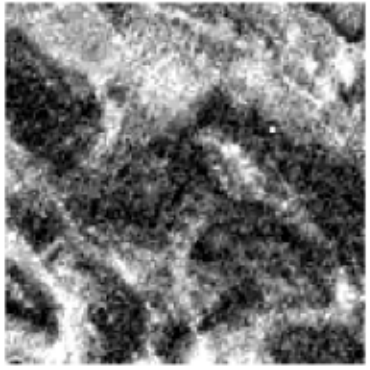


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_A_B02

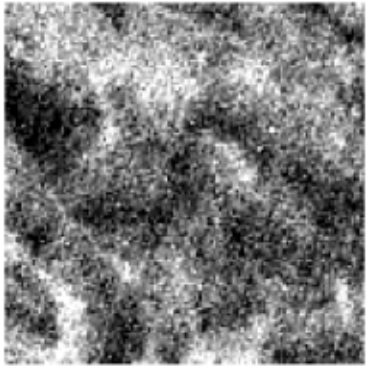


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_B1_B03

原生真值



可见 B02

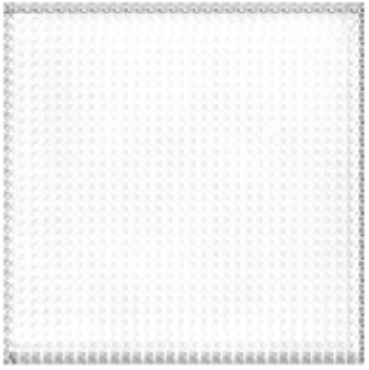


目标掩码

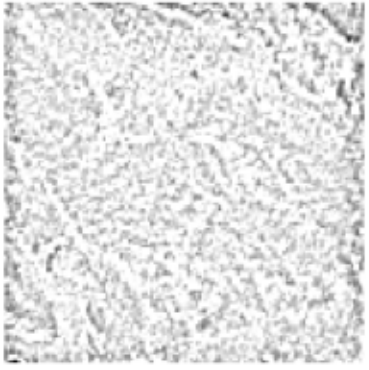
FiLM 浅层 ResNet
+ D0 连续插值解码
共享主路径输出



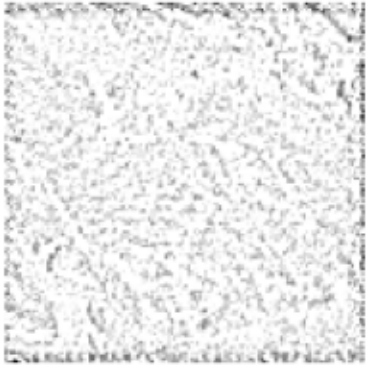
FiLM 浅层 ResNet
+ D1 子像素解码
共享主路径输出



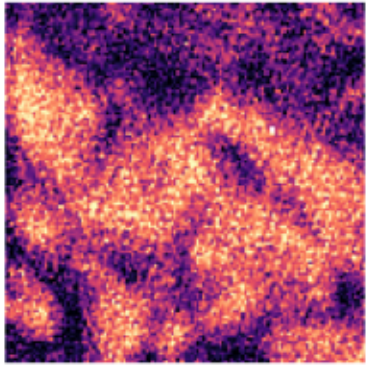
FiLM 浅层 ResNet
+ D0 连续插值解码
叠加历史支路后的总输出



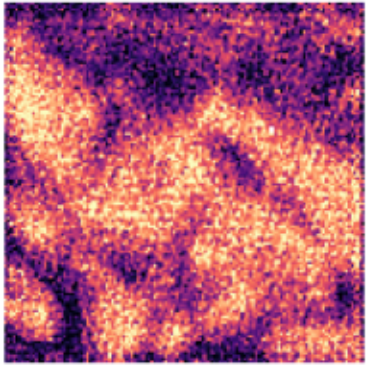
FiLM 浅层 ResNet
+ D1 子像素解码
叠加历史支路后的总输出



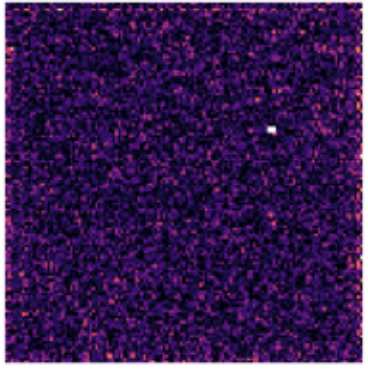
FiLM 浅层 ResNet
+ D0 连续插值解码
绝对误差



FiLM 浅层 ResNet
+ D1 子像素解码
绝对误差



候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_A_SR_B1

原生真值



可见输入



目标掩码



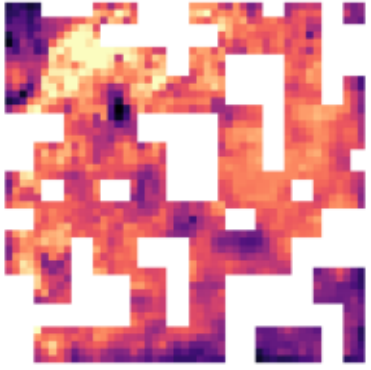
FiLM 浅层 ResNet
+ D0 连续插值解码
共享主路径输出

FiLM 浅层 ResNet
+ D1 子像素解码
共享主路径输出

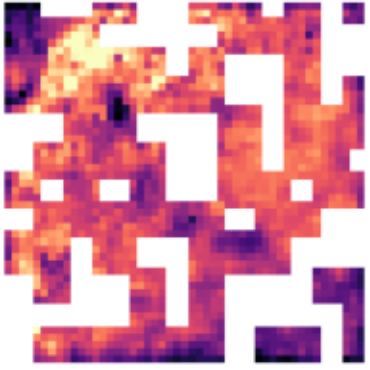
FiLM 浅层 ResNet
+ D0 连续插值解码
总输出

FiLM 浅层 ResNet
+ D1 子像素解码
总输出

FiLM 浅层 ResNet
+ D0 连续插值解码
绝对误差



FiLM 浅层 ResNet
+ D1 子像素解码
绝对误差



候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_B1_SR_B2

原生真值



可见 SR_B1

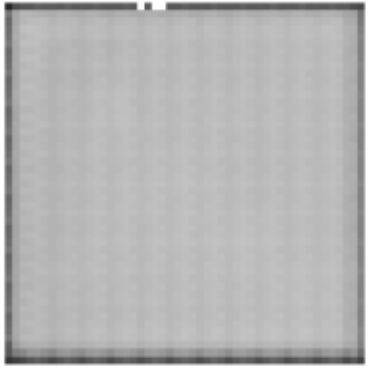


目标掩码

FiLM 浅层 ResNet
+ D0 连续插值解码
共享主路径输出



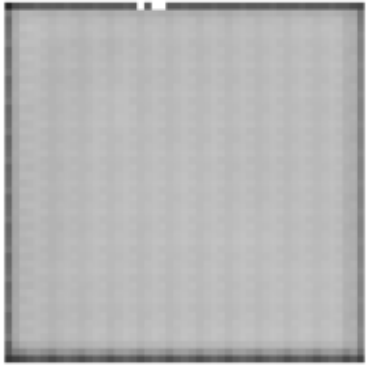
FiLM 浅层 ResNet
+ D1 子像素解码
共享主路径输出



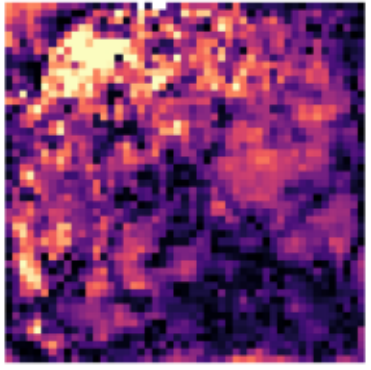
FiLM 浅层 ResNet
+ D0 连续插值解码
叠加历史支路后的总输出



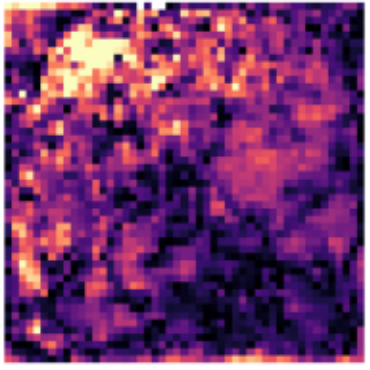
FiLM 浅层 ResNet
+ D1 子像素解码
叠加历史支路后的总输出



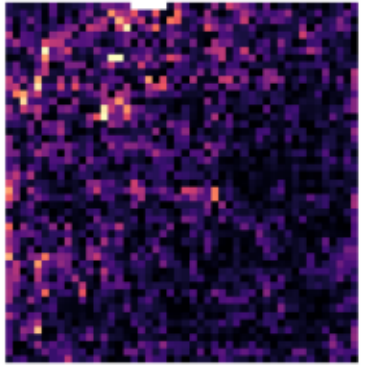
FiLM 浅层 ResNet
+ D0 连续插值解码
绝对误差



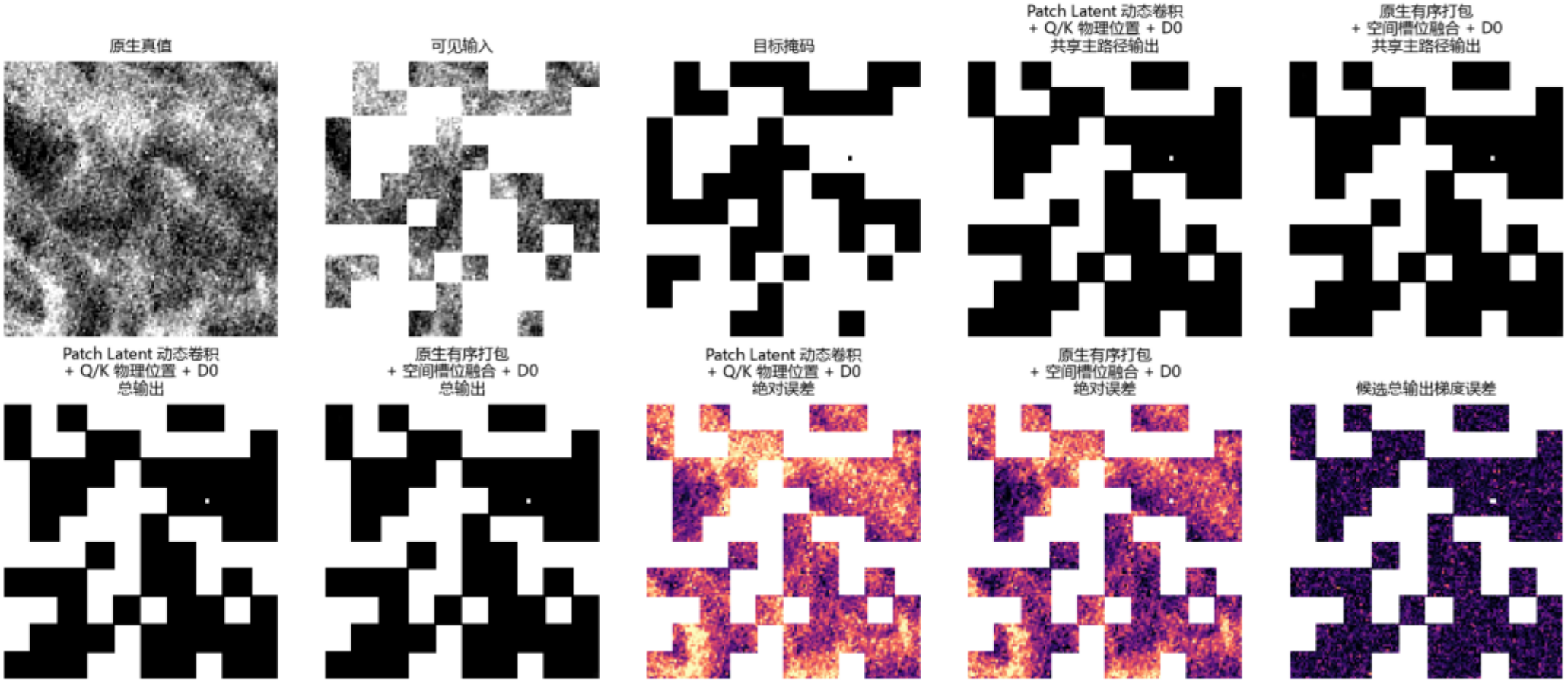
FiLM 浅层 ResNet
+ D1 子像素解码
绝对误差



候选总输出梯度误差

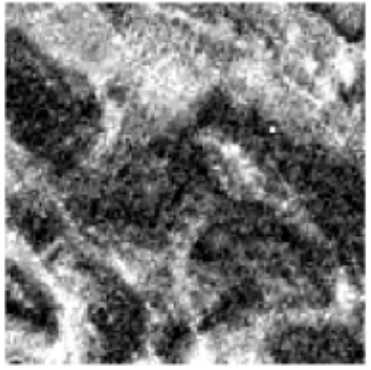


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_A_B02

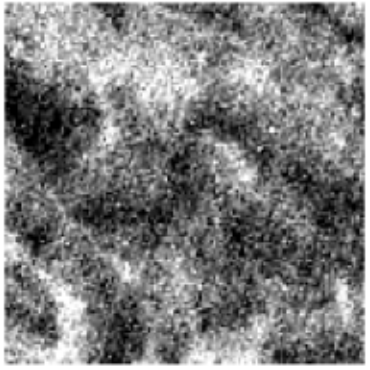


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_B1_B03

原生真值



可见 B02



目标掩码

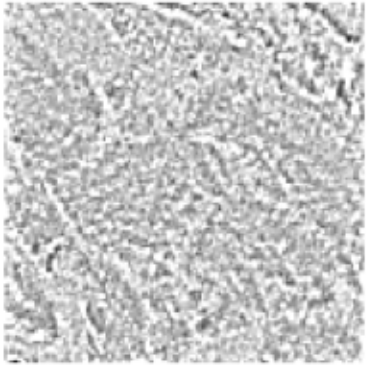
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
共享主路径输出



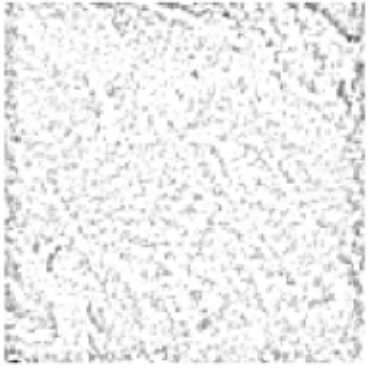
原生有序打包
+ 空间槽位融合 + D0
共享主路径输出



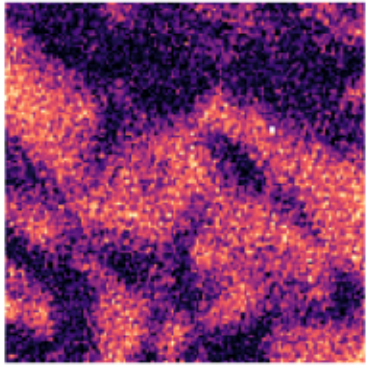
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
叠加历史支路后的总输出



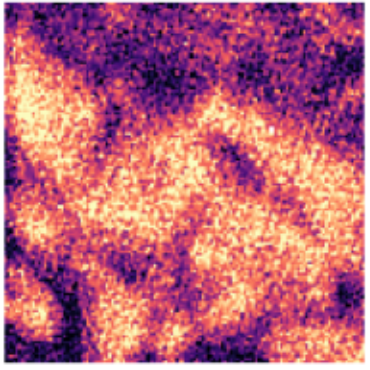
原生有序打包
+ 空间槽位融合 + D0
叠加历史支路后的总输出



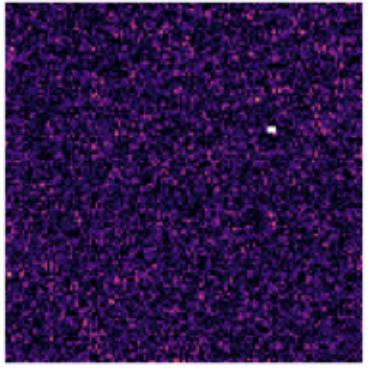
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
绝对误差



原生有序打包
+ 空间槽位融合 + D0
绝对误差

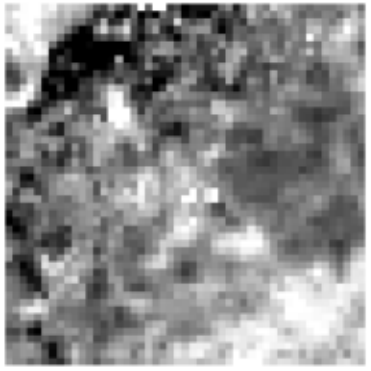


候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_A_SR_B1

原生真值



可见输入



目标掩码



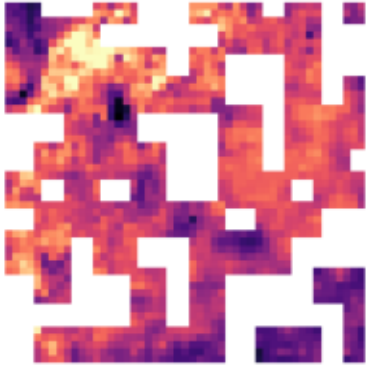
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
共享主路径输出

原生有序打包
+ 空间槽位融合 + D0
共享主路径输出

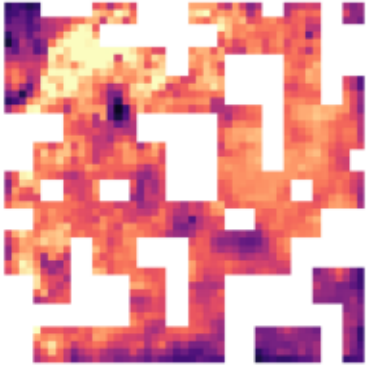
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
总输出

原生有序打包
+ 空间槽位融合 + D0
总输出

Patch Latent 动态卷积
+ Q/K 物理位置 + D0
绝对误差



原生有序打包
+ 空间槽位融合 + D0
绝对误差



候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_B1_SR_B2

原生真值



可见 SR_B1



目标掩码

Patch Latent 动态卷积
+ Q/K 物理位置 + D0
共享主路径输出



原生有序打包
+ 空间槽位融合 + D0
共享主路径输出



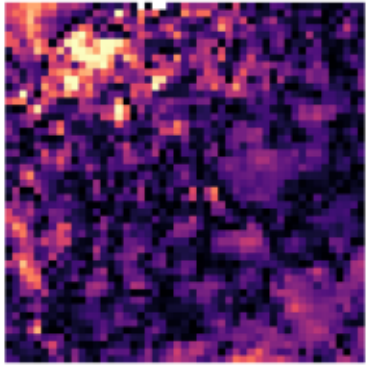
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
叠加历史支路后的总输出



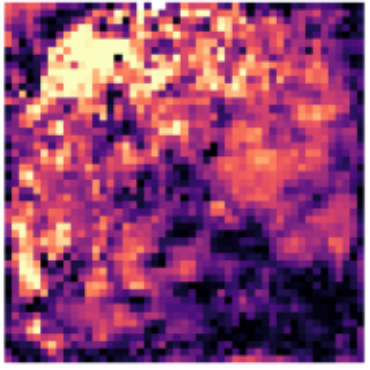
原生有序打包
+ 空间槽位融合 + D0
叠加历史支路后的总输出



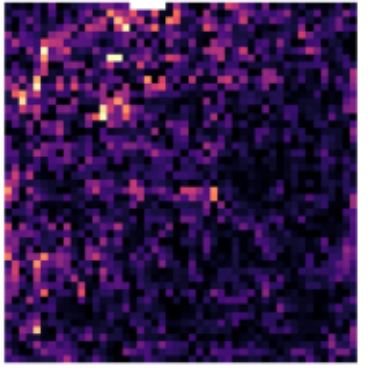
Patch Latent 动态卷积
+ Q/K 物理位置 + D0
绝对误差



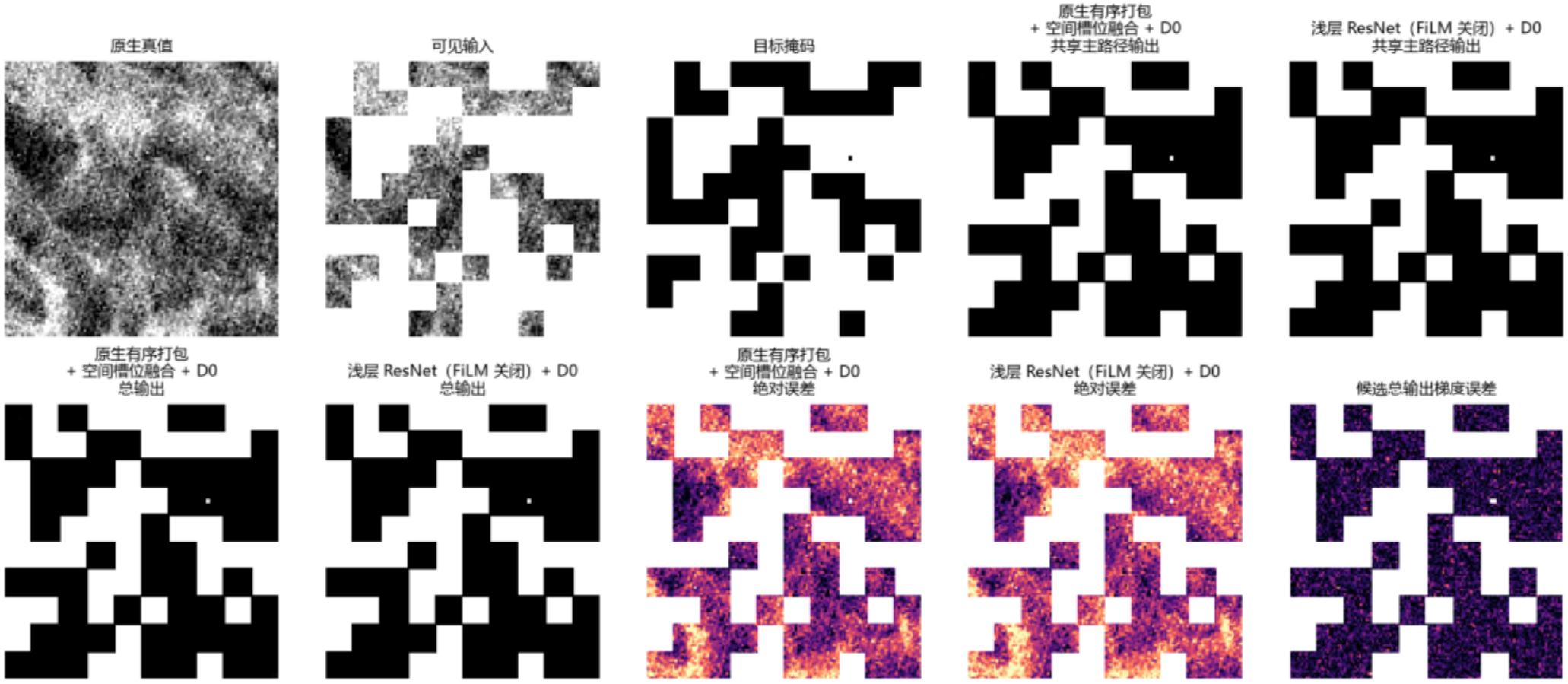
原生有序打包
+ 空间槽位融合 + D0
绝对误差



候选总输出梯度误差

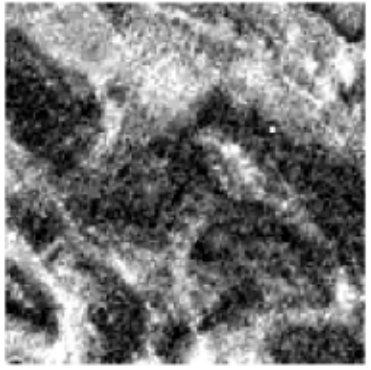


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_A_B02

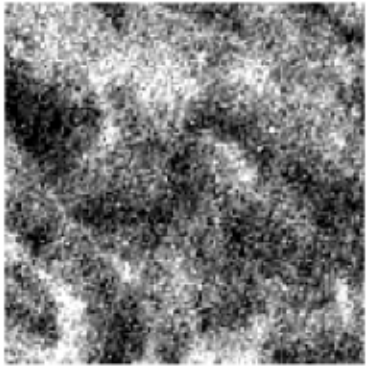


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_B1_B03

原生真值



可见 B02



目标掩码

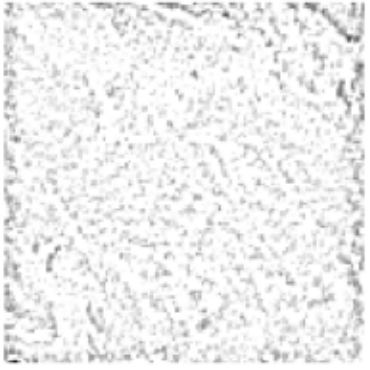
原生有序打包
+ 空间槽位融合 + D0
共享主路径输出



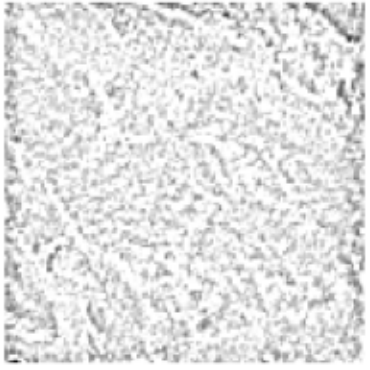
浅层 ResNet（FiLM 关闭）+ D0
共享主路径输出



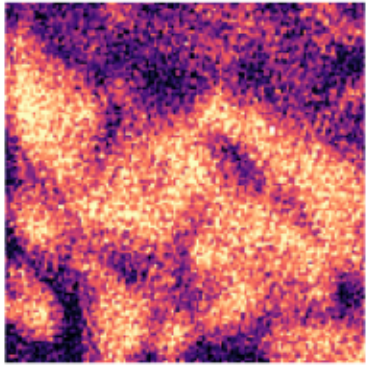
原生有序打包
+ 空间槽位融合 + D0
叠加历史支路后的总输出



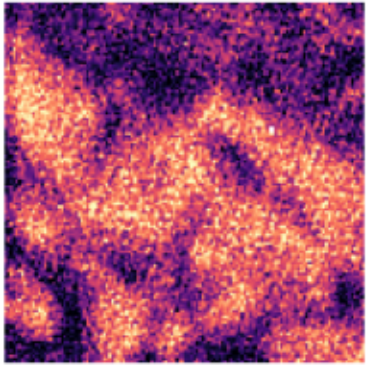
浅层 ResNet（FiLM 关闭）+ D0
叠加历史支路后的总输出



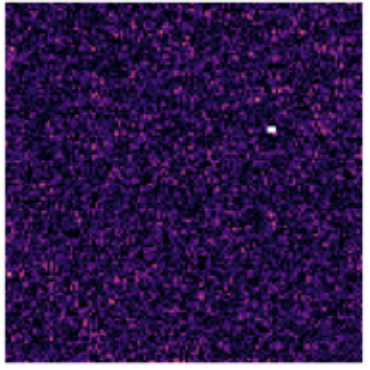
原生有序打包
+ 空间槽位融合 + D0
绝对误差



浅层 ResNet（FiLM 关闭）+ D0
绝对误差

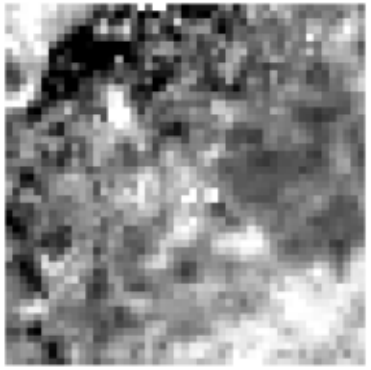


候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_A_SR_B1

原生真值



可见输入



目标掩码



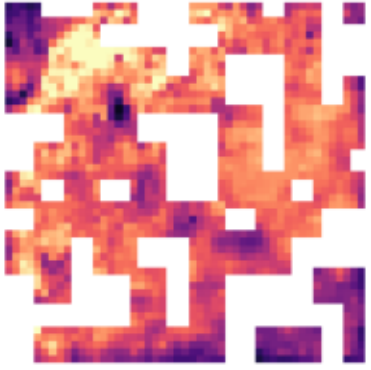
原生有序打包
+ 空间槽位融合 + D0
共享主路径输出

浅层 ResNet（FiLM 关闭）+ D0
共享主路径输出

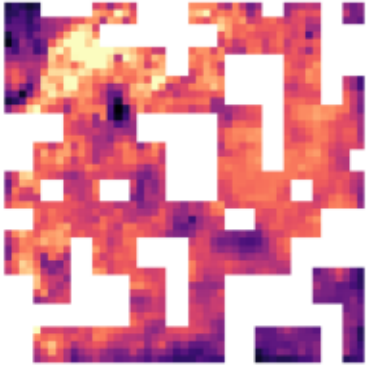
原生有序打包
+ 空间槽位融合 + D0
总输出

浅层 ResNet（FiLM 关闭）+ D0
总输出

原生有序打包
+ 空间槽位融合 + D0
绝对误差



浅层 ResNet（FiLM 关闭）+ D0
绝对误差



候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_B1_SR_B2

原生真值



可见 SR_B1



目标掩码



原生有序打包
+ 空间槽位融合 + D0
共享主路径输出



浅层 ResNet（FiLM 关闭）+ D0
共享主路径输出



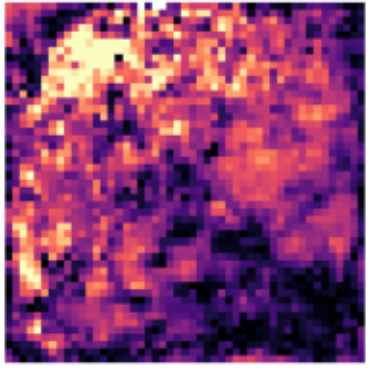
原生有序打包
+ 空间槽位融合 + D0
叠加历史支路后的总输出



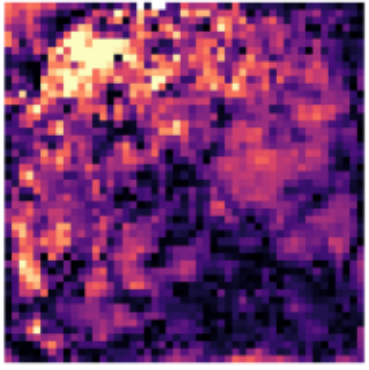
浅层 ResNet（FiLM 关闭）+ D0
叠加历史支路后的总输出



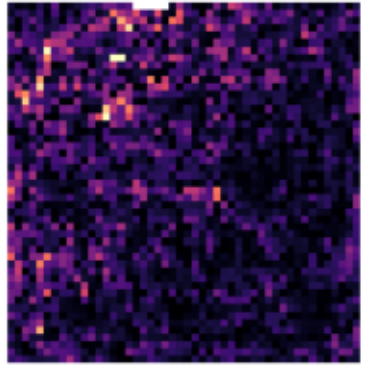
原生有序打包
+ 空间槽位融合 + D0
绝对误差



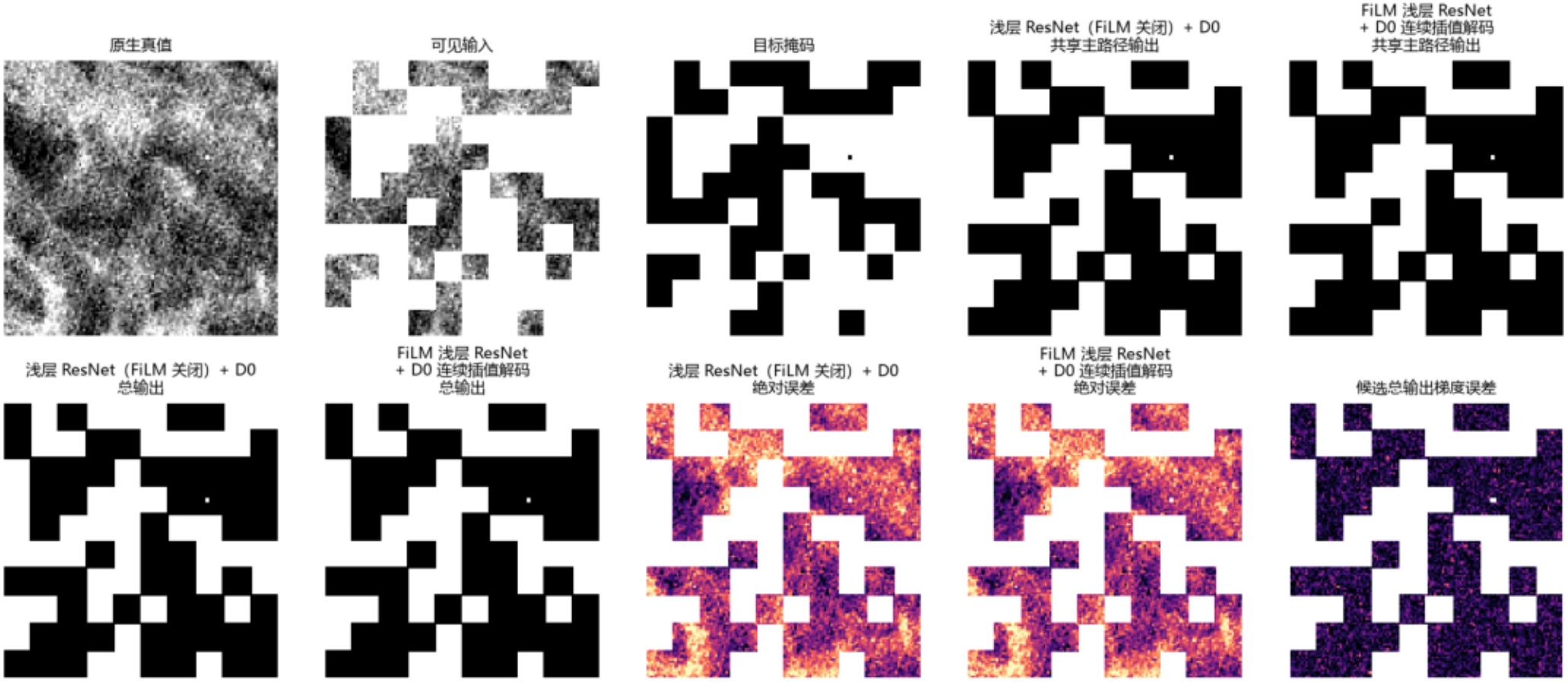
浅层 ResNet（FiLM 关闭）+ D0
绝对误差



候选总输出梯度误差

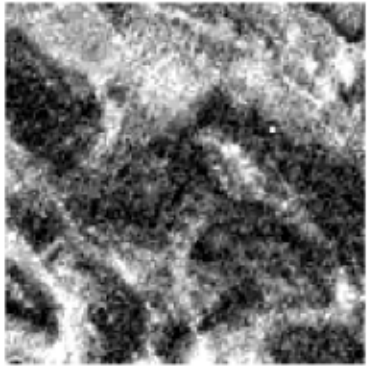


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_A_B02

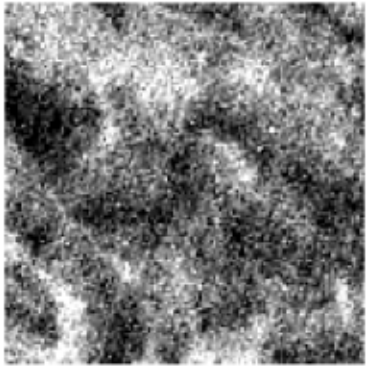


V3-FR · Step 233 · obs_5929ceec240fa570c7a1a784_B1_B03

原生真值



可见 B02



目标掩码

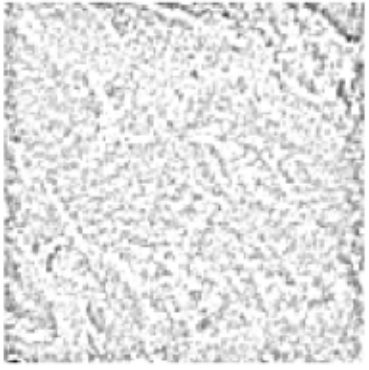
浅层 ResNet（FiLM 关闭）+ D0
共享主路径输出



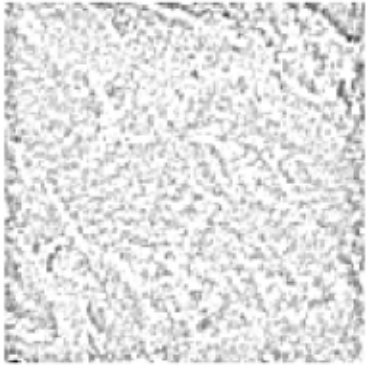
FiLM 浅层 ResNet
+ D0 连续插值解码
共享主路径输出



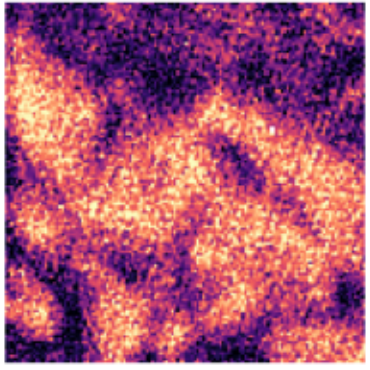
浅层 ResNet（FiLM 关闭）+ D0
叠加历史支路后的总输出



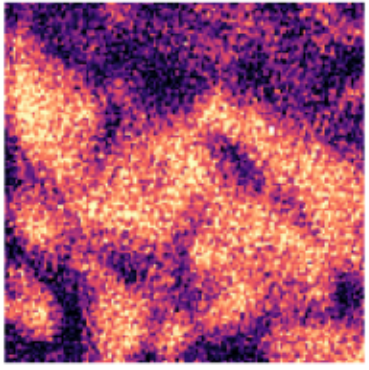
FiLM 浅层 ResNet
+ D0 连续插值解码
叠加历史支路后的总输出



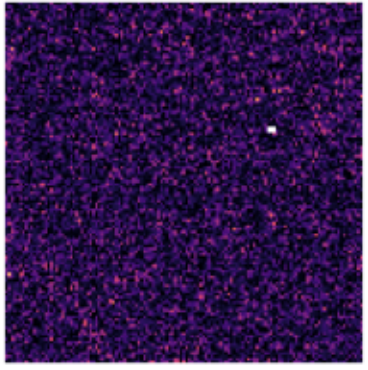
浅层 ResNet（FiLM 关闭）+ D0
绝对误差



FiLM 浅层 ResNet
+ D0 连续插值解码
绝对误差

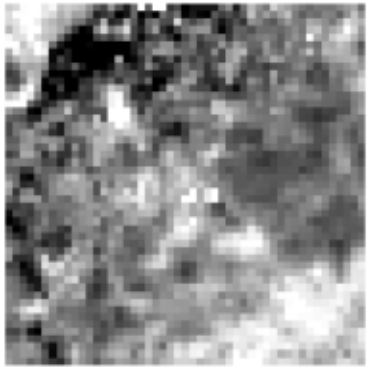


候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_A_SR_B1

原生真值



可见输入



目标掩码



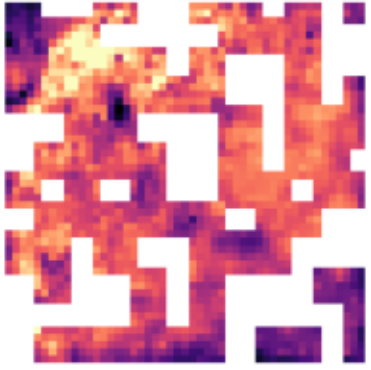
浅层 ResNet（FiLM 关闭）+ D0
共享主路径输出

FiLM 浅层 ResNet
+ D0 连续插值解码
共享主路径输出

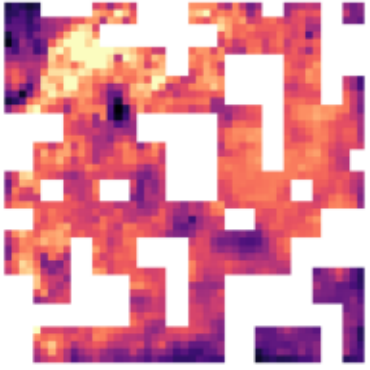
浅层 ResNet（FiLM 关闭）+ D0
总输出

FiLM 浅层 ResNet
+ D0 连续插值解码
总输出

浅层 ResNet（FiLM 关闭）+ D0
绝对误差



FiLM 浅层 ResNet
+ D0 连续插值解码
绝对误差



候选总输出梯度误差



V3-FR · Step 233 · obs_64d10dd286169cf55d8668ac_B1_SR_B2

原生真值



可见 SR_B1



目标掩码

浅层 ResNet（FiLM 关闭）+ D0
共享主路径输出



FiLM 浅层 ResNet
+ D0 连续插值解码
共享主路径输出



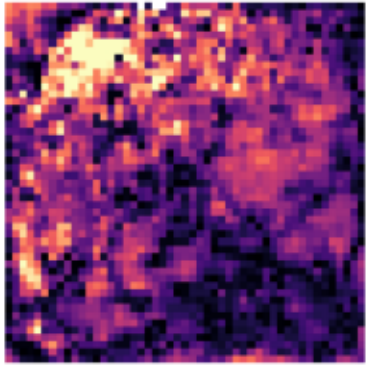
浅层 ResNet（FiLM 关闭）+ D0
叠加历史支路后的总输出



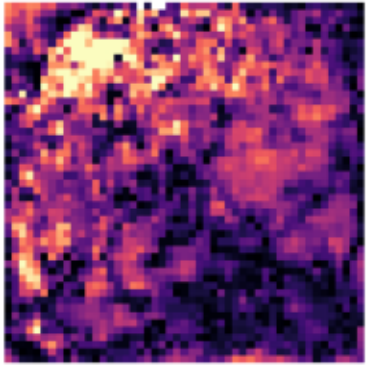
FiLM 浅层 ResNet
+ D0 连续插值解码
叠加历史支路后的总输出



浅层 ResNet（FiLM 关闭）+ D0
绝对误差



FiLM 浅层 ResNet
+ D0 连续插值解码
绝对误差



候选总输出梯度误差

